RpS12-mediated induction of the *Xrp1*^{short} isoform links ribosomal protein mutations to cell competition

Eleni Tsakiri[1,5,6], Myrto Potiri[1,3,5], Kyriaki Kontogiannidi[1,4,5], Katerina Douka[1], Kiriaki Kanakousaki[1], Nikoleta Vavouraki[1], Maria Loizou[1], Panagiotis Moulos[1], Efthimios M.C. Skoulakis[1], Martina Samiotaki[2], Panagiota Kafasla[1], Marianthi Kiparaki[1,7,]*.

[1]Institute for Fundamental Biomedical Research, B.S.R.C. “Alexander Fleming”, 16672, Vari, Attica, Greece

[2]Institute for Bioinnovation, B.S.R.C. “Alexander Fleming”, 16672, Vari, Attica, Greece

[3]Department of Biochemistry and Biotechnology, University of Thessaly, 41500, Larissa, Greece.

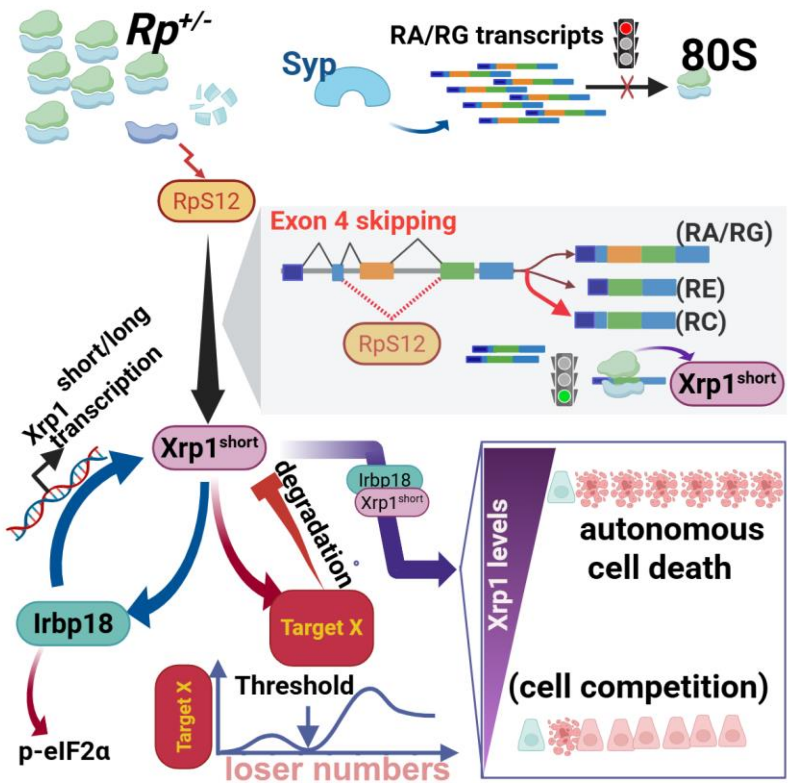
[4] Medical School, National and Kapodistrian University of Athens, 11527, Athens, Greece.

[5] These authors contributed equally

[6] Present address: Department of Physiology, Medical School, National and Kapodistrian University of Athens, 11527, Athens, Greece

[7]Lead Contact

*Correspondence: kiparaki@fleming.gr



Highlights

1. RpS12-dependent Xrp1^{short} splicing initiates loser signaling in ribosomal protein mutants
2. RpS12 or Xrp1 overexpression confers loser status on wild-type cells
3. Syncrip modulates Xrp1^{short} production; its depletion induces Xrp1-driven cell competition
4. Post-transcriptional regulation of Xrp1 mediates context-dependent loser cell adaptation

eTOC blurb

Tsakiri et al. show that RpS12 promotes the Xrp1^{short} splice isoform, initiating loser signaling in ribosomal mutants. RpS12 or Xrp1 overexpression is sufficient to drive cell competition, while Syncrip represses Xrp1. Context-dependent post-transcriptional control of Xrp1 helps determine whether cells die autonomously, are eliminated by neighbors, or adapt.

SUMMARY

Cell competition, the context–dependent cell elimination phenomenon via short-range cell–cell interactions, was originally described in *Drosophila*, where heterozygous ribosomal protein *(Rp*$^{+/-}$) mutant cells are outcompeted by wild-type neighbors. The transcription factor Xrp1 mediates most *Rp*$^{+/-}$ phenotypes, including reduced translation and competitiveness, yet how RpS12 induces Xrp1 was unclear. We show that RpS12 promotes a splice variant of *Xrp1* encoding the Xrp1^{short} isoform, which defines the *Rp*$^{+/-}$ loser identity. RpS12 overexpression is sufficient to induce *Xrp1*^{short} and confer loser status. Expression of either Xrp1^{short} or Xrp1^{long} isoform can trigger competition, establishing *Xrp1* as the central driver of the loser fate. When the number of losers exceeds a critical threshold, a quorum-like response supports their survival by post-transcriptionally downregulating Xrp1. We identify the RNA-binding protein Syncrip, reduced in *Rp*$^{+/-}$ cells, as an *Xrp1* regulator whose depletion activates Xrp1^{short}-dependent competition. Therefore, RpS12-dependent *Xrp1*^{short} expression emerges as the primary *Rp*$^{+/-}$ signal initiating cell competition.

Keywords:

Drosophila development; cell competition; minute mutation; ribosomal protein;

ribosomopathy; splicing; Xrp1; RNA-binding proteins; Syncrip/hnRNPQ; quorum-like response.

INTRODUCTION

Life is inextricably linked to the function of ribosomal protein (*Rp*) genes, as most homozygous *Rp* mutations are lethal. In humans, heterozygous *Rp* mutations (*Rp*$^{+/−}$) cause Diamond–Blackfan anemia [1,2], while in *Drosophila* they produce the *Minute* syndrome, characterized by developmental delay, slow growth, and short bristles[3].

In *Drosophila* mosaic imaginal discs (rapidly growing larval epithelial tissues that generate adult structures), *Rp*$^{+/−}$ cells (called "losers") are eliminated by wild-type neighbors (called "winners") through apoptosis at clone boundaries[4-8]. This boundary cell death is a major hallmark of *Rp*$^{+/−}$-driven cell competition and reflects its short-range nature[9]. Cell competition has been identified in *Drosophila* and mammals during development, tissue homeostasis, ageing, and disease, including cancer[8,10-14]. Nevertheless, the mechanisms driving cell competition remain incompletely understood.

Xrp1 is a rapidly evolving AT-hook/bZIP transcription factor that mediates most *Rp*$^{+/−}$ responses in *Drosophila,* including reduced competitiveness, slower growth, altered gene expression, developmental delay, and decreased translation[7,15-17]. Xrp1 forms a heterodimer with Irbp18, and both are upregulated in *Rp*$^{+/−}$ cells through autoregulation[15,16,18-20]. Heterozygous *Xrp1*-null mutations (*Xrp1*$^{+/−}$) are haploinsufficient, preventing Xrp1 induction and rescuing *Rp*$^{+/−}$ phenotypes, including cell competition[15,19,21].

RpS12, a non-*Minute Rp*, is required for Xrp1 activation in *Rp*$^{+/−}$ cells[3,15,21,22]. *Rp*$^{+/−}$ cells homozygous for the *RpS12*^{G97D} mutation, substituting glycine 97 with aspartate, fail to activate Xrp1[15,21-23]. G97-dependent activity is distinct from the gene's essential role, as *RpS12*^{G97D} homozygotes are viable[21,22]. Although increasing RpS12 levels dramatically reduces growth of *Rp*$^{+/−}$ cells, the effect is modest in wild-type cells[22], suggesting that RpS12 requires an additional *Rp*$^{+/−}$-specific signal activate Xrp1 and initiate competition[15,21,22].

Xrp1 also functions as a key stress sensor and effector, responding to diverse stresses[7,19,24-30]. In addition, the RpS12–Xrp1 pathway coordinates inter-organ growth through Dilp8[31] and eliminates certain aneuploid cells during development[32,33]. Understanding how RpS12 activates Xrp1 is therefore critical for deciphering ribosome-linked homeostatic mechanisms.

It remains unclear whether RpS12 is the primary signal inducing Xrp1 in *Rp*$^{+/-}$ cells, or whether additional *Rp*$^{+/-}$-specific signals are required, and whether Xrp1 alone can initiate cell competition. Our findings reveal the mechanism of Xrp1 activation and explain how *Rp* mutations drive cell competition.

RESULTS

RpS12 overexpression is sufficient to induce Xrp1-dependent cell competition, independently of proteostasis overload

In *Rp*$^{+/-}$ discs, clones carrying two extra copies of the *RpS12* genomic locus (4x*RpS12*$^{+}$) are completely eliminated, whereas their twin spots with wild-type *RpS12* levels (2x*RpS12*$^{+}$) survive[22]. In wild-type discs, however, 4x*RpS12*$^{+}$ clones show a modest yet significant 24% reduction in size relative to 2x*RpS12*$^{+}$ twin-spots, suggesting that RpS12 requires an *Rp*$^{+/-}$-specific signal to activate *Xrp1*-dependent competition[15,21,22].

However, we found that RpS12 overexpression in wild-type wing disc clones was sufficient to induce Xrp1, whereas RpS12^{G97D} or GFP overexpression did not (Figures 1A, 1B, and S1A). RpS12-overexpressing clones exhibited Xrp1-dependent apoptotic cell death, predominantly at boundaries with wild-type cells (Figures 1C−E, 1I, S1L, and S1M, quantified in 1J).

Because proteotoxic stress caused by unassembled ribosomal proteins (RPs) has been proposed as the initiating trigger for *Rp*$^{+/-}$ responses in both *Drosophila* and yeast[24,34-37], we tested whether generic RP overexpression could reproduce the RpS12 phenotype. Overexpression of RpS12^{G97D}, RpL14, RpL13 or RpS17 failed to robustly induce Xrp1 (Figures S1B−S1G) or trigger cell death (Figures 1F−1H, and S1H). GFP-RpL10Ab overexpression, using a transgene employed for cell-specific translational profiling[38], yielded no clones and strongly induced Xrp1 while reducing compartment size (Figures S1I−S1K), although the effect of untagged RpL10Ab remains to be determined. Therefore, RpS12-driven competition is highly specific, requiring Gly97 and Xrp1 induction. In addition, RpS12 overexpression in the pouch using nub-Gal4 caused sporadic cell death (Figures S1N, control as in S5B), resembling the autonomous Xrp1-dependent death found in $Rp^{+/-}$ cells [5,35,39,40].

In $Rp^{+/-}$ cells, RpS12 induces the Xrp1 RC splicing variant, which encodes the Xrp1^{short} protein isoform.

The *Xrp1* locus produces four long (*RA, RB, RF, RG*) and three short (*RC, RD, RE*) mRNA isoforms through differential promoter usage and splicing, encoding the Xrp1^{long} (73 kDa) and Xrp1^{short} (45 kDa) proteins, respectively (Figure 2A)[41]. Both isoforms share the C-terminal AT-hook and b-ZIP domains but differ at their N-termini, with translation initiating from exon 4 in Xrp1^{long} and exon 5 in Xrp1^{short} (Figure 2A).

Xrp1 mRNA is bound by multiple RNA-binding proteins (RBPs) [42], and is efficiently co-transcriptionally spliced [43]. To examine Xrp1 alternative splicing (AS), we analyzed published RNA-seq datasets from seven genotypes (a) wild-type, (b) $RpS17^{+/-}$, (c) $RpS3^{+/-}$,(d) $Xrp1^{+/-}$, $RpS3^{+/-}$, (e) $Xrp1^{+/-}$, (f) $RpS12^{G97D/G97D}$, and (g) $RpS12^{G97D/G97D}$, $RpS3^{+/-}$[15,21] (Figures S2A-S2B). These datasets use the dominant-null $Xrp1^{M2\text{-}73}$ allele, which carries a stop codon in exon 5 and does not produce detectable protein (Figure 3A and 3B). $Xrp1^{M2\text{-}73/+}$ heterozygosity (depicted as $Xrp1^{+/-}$) prevents Xrp1 activation in $Rp^{+/-}$ cells and rescues *Minute* phenotypes, including cell competition[15,19,21].

Percent Spliced In (PSI, Ψ) values, representing the fraction of mRNA reads that including a given AS event, were used to quantify alternative splicing. Principal component analysis (PCA) of PSI values for alternative exon events separated $RpS12^{G97D}$ genotypes from other samples, indicating RpS12-dependent splicing changes (Figure 2B). $Rp^{+/-}$ mutants ($RpS3^{+/-}$ and $RpS17^{+/-}$) clustered separately along PC2, suggesting additional AS changes that require both RpS12 and Xrp1 function (Figure 2B). PCA of all AS events likewise separated $RpS12^{G97D}$ genotypes along PC1 (Figure S2D).

Analysis showed that Xrp1 exon-4 was highly included in all samples (PSI values ~92-99%) (Figures 2D, S2C, S2E and Supplementary Table S1). $Rp^{+/-}$ mutants exhibited increased exon-4 skipping (8% vs 2% in wild-type), and this was also present in *Xrp1*$^{+/-}$, *RpS3*$^{+/-}$cells. In contrast, *RpS12*$^{G97D/G97D}$, *RpS3*$^{+/-}$ cells reverted exon-4 inclusion to wild-type or lower levels, indicating an RpS12-dependent mechanism of *Xrp1* splicing, as visualized by Sashimi plots (Figures 2D-2E).

Semi-quantitative RT-PCR confirmed increased exon-4 skipping in both *Rp*$^{+/-}$ mutants compared with wild-type. Increased exon 4 skipping persisted in the presence of the heterozygous *Xrp1*$^{M2\text{-}73}$ mutation, but was reduced by the *RpS12*^{G97D} mutation in both *Rp*$^{+/-}$ and wild-type cells (Figure 2F). Analysis of Xrp1 isoform in the same RNA-seq datasets[15,21] showed that the long *RA/ RG* isoforms, which differ only in their 3' untranslated regions (UTR), are the most abundant and are approximately 2-fold upregulated in *Rp*$^{+/-}$ mutants in an Xrp1- and RpS12-dependent manner, consistent with transcriptional autoregulation (Figures 2G and S2F)[15,16,18,21]. By contrast, the less abundant *Xrp1-RC* isoform, which is also transcribed from the *Xrp1-RA/ RG* promoter (Figure 2A), showed the strongest induction (~13-fold) in *Rp*$^{+/-}$ cells. This induuction was completely dependent on RpS12 and only partially dependent on Xrp1 (Figures 2G and S2F). Because PSI measures relative splice choice, isoform analysis uncovers the additional transcriptional autoregulation, explaining the strong *Xrp1-RC* induction despite modest PSI changes. Collectively, these data indicate that in *Rp*$^{+/-}$ cells, RpS12 primarily induces the *Xrp1-RC* transcript, while *Xrp1* autoregulation further increases the expression of all *Xrp1* transcripts (Figure 2C).

In *Rp*$^{+/-}$ cells, the Xrp1^{short} isoform is required for cell competition, while Xrp1^{long} is dispensable for RpS12-dependent induction of Xrp1^{short}.

Upstream open reading frames (uORFs) can repress translation of downstream ORFs in stress-responsive genes, including p-eIF2α-regulated transcripts[44]. Exon 4 contains uORFs[28,29], which may explain why Xrp1^{long} protein is barely detectable, despite *Xrp1-RA/RG* being the most abundant transcripts, whereas the low-abundance RC transcript robustly produces Xrp1^{short} in *Rp*$^{+/-}$ cells (Figures 3A, 3B, S3C and S3D).

The *Xrp1*^{08} allele, which blocks *Rp*$^{+/-}$ competitive elimination, disrupts a splice enhancer near exon 4[16] (Figure S3A). In wild-type cells, Xrp1 shows basal autoregulatory activity, as *Xrp1*-lacZ expression requires both Xrp1 and Irbp18 [24]. *Xrp1*$^{08/08}$ and *RpS12*$^{G97/G97D}$ cells show reduced levels of *Xrp1-RC* short transcript without affecting the abundant *Xrp1-RA/RG* long transcripts (Figures 2F, 3C, 3D, S2F and Suppl. Data from Baillon et al[16]), suggesting that Xrp1$^{\mathrm{long}}$ drives basal autoregulation in wild-type cells. As expected, in *Xrp1*$^{08/08}$,*RpS3*$^{+/-}$ and *RpS12*$^{G97D/G97D}$,*RpS3*$^{+/-}$ cells, exon-4 skipping and RC induction are not observed(Figures 2D, 2F, 2G, 3D, and 3G), and Xrp1$^{\mathrm{short}}$ protein is not induced (Figure 3B). However, since *Xrp1*$^{\mathrm{long}}$ transcripts are also not upregulated in these cells (Figures 2G, 3C, 3F, and S2F), this suggests that Xrp1$^{\mathrm{short}}$ is required for *Rp*$^{+/-}$-dependent autoregulation.

We confirmed that, *Xrp1*$^{M2\text{-}73/+}$, *RpS3*$^{+/-}$ and *Xrp1*$^{M2\text{-}73/+}$, *RpS17*$^{+/-}$ cells do not induce Xrp1$^{\mathrm{short}}$ protein (Figures 3B and S3C) as they do not activate the *Rp*$^{+/-}$ pathway[15,19,21]. However, in *Xrp1*$^{M2\text{-}73/+}$, *RpS17*$^{+/-}$ cells, Xrp1$^{\mathrm{long}}$ exceeds its wild-type levels (Figure 3B).

To exclude the possibility that RpS12 promotes *Xrp1-RC* indirectly through uORF-mediated translation of Xrp1$^{\mathrm{long}}$, we quantified long (*RA/RB/RF/RG*) and short (*RC*) transcripts in (1) *wild-type*, (2) *RpS3*$^{+/-}$, (3) *RpS12*$^{G97D/G97D}$; *RpS3*$^{+/-}$, (4) *Xrp1*$^{M2\text{-}73/+}$, *RpS3*$^{+/-}$, and (5) *Xrp1*$^{M2\text{-}73/M2\text{-}73}$, *RpS3*$^{+/-}$ cells. Importantly, the M2-73 mutation does not alter mRNA abundance or stability (Figure S3B), despite Xrp1 being a nonsense-mediated decay (NMD) target in *Drosophila*[45].

Indeed, Xrp1$^{\mathrm{long}}$ transcripts are not upregulated in *Xrp1*$^{M2\text{-}73/+}$*, RpS3*$^{+/-}$ and *RpS12*$^{G97D/G97D}$*, RpS3*$^{+/-}$ cells (Figure 3F), consistent with a requirement for both *Xrp1* locus and for RpS12-G97 function in *Rp*-dependent Xrp1 autoregulation. In *Xrp1*$^{M2\text{-}73/\ M2\text{-}73}$*, RpS3*$^{+/-}$ cells, long transcripts fell to 50% of wild-type (Figure 3F), indicating that one Xrp1 locus is sufficient for basal transcription.

Importantly, *Xrp1*$^{M2\text{-}73/\ M2\text{-}73}$*, RpS3*$^{+/-}$ cells still showed elevated *Xrp1-RC* short transcript relative to wild-type (Figure 3G), despite loss of Xrp1 protein and autoregulation. Indeed, the abundance of the *Xrp1-RC* transcript relative to all long transcripts is increased in *Rp*$^{+/-}$ cells compared with wild-type and reduced by *RpS12*$^{G97D/G97D}$ and *Xrp1*$^{08/08}$ (Figures 3E and 3H). Importantly, this ratio is further elevated in the complete absence of Xrp1 (Figure 3H). Therefore, in *Xrp1*$^{M2\text{-}73/M2\text{-}73}$*, RpS3*$^{+/-}$ cells, RpS12 can still induce *Xrp1-RC* transcript independently of prior translation of Xrp1$^{\mathrm{long}}$. Additionally, similar to *RpS18*$^{+/-}$ cells[19], clonal PPP1R15 overexpression (the PP1 regulatory subunit that dephosphorylates eIF2α) completely abolished p-eIF2α levels in *RpS17*$^{+/-}$ cells without reducing Xrp1, whereas Irbp18 depletion reduced both Xrp1 and p-eIF2α levels (Figures S3E- S3G).

Altogether, these data support a model in which, in *Rp*$^{+/-}$ cells, RpS12-dependent induction of Xrp1^{short} through exon 4 skipping is the primary driver of cell competition, independent of uORF-mediated translational regulation, whereas p-eIF2α elevation appears downstream of Irbp18/Xrp1^{short} activity (Figures 2-4, S3).

RpS12 is sufficient to induce the short splicing variant in wild-type cells and to induce Xrp1^{short} protein in a dosage dependent manner.

Semi-quantitative RT-PCR showed that RpS12 overexpression in wild-type cells reproduced the exon-4 skipping observed in *Rp*$^{+/-}$ mutants (Figure 4A and 2F). qRT-PCR further showed that RpS12 selectively increased Xrp1^{short} without affecting Xrp1^{long}, whereas RpS12^{G97D} had no effect (Figures 4A and 4B), consistent with the requirement for Gly97 (Figures 1-3).

Despite inducing Xrp1-HA expression (Figures 4D; control similar to S1A), RpS12 overexpression did not induce *Xrp1*-lacZ (Figures 4C; control S4A) or p-eIF2α (Figures 4E and 4F; control S4C), suggesting that the Xrp1^{short} levels produced were insufficient to trigger *Rp*$^{+/-}$ autoregulation. In contrast, TAF1B depletion strongly induced *Xrp1*-lacZ (Figure S4B), consistent with reported Xrp1 and p-eIF2α upregulation[19]. Comparative staining showed lower and more heterogeneous Xrp1 levels in RpS12-overexpressing clones than in *Rp*$^{+/-}$ cells, and even lower in discs ubiquitously expressing UAS-RpS12 under actin-Gal4 (act>RpS12, same driver as used in RT-PCR) (Figures S4D-S4H, controls S1A, S4I). These lower Xrp1 levels, together with the lack of transcriptional autoregulation, likely explain why RpS12 overexpression did not yield the same *Xrp1-RC* increase observed in *Rp*$^{+/-}$ cells. Indeed, in Xrp1$^{+/-}$, RpS3$^{+/-}$ cells, where autoregulation is lost but RpS12 activity is present, *Xrp1-RC* induction is reduced from 13fold in RpS3$^{+/-}$ cells to 5-fold, becoming similar to that seen in RpS12-overexpressing levels (5-fold versus 2.5-fold induction) (Figure S2).

Increasing RpS12 dosage by using two copies of the UAS-RpS12 transgene (2xRpS12) enhanced Xrp1 induction and markedly increased cell death (Figures 4G-4J, S4J, S4K). Together, these data show that RpS12 acts as a dose-dependent upstream regulator of *Xrp1* splicing and loser status.

Overexpression of either Xrp1 isoform is sufficient to transform wild-type cells into losers

Although Xrp1's role in cell competition is established[7], whether Xrp1 alone can initiate this process has remained unclear because of its high toxicity. Overexpression of either Xrp1 isoform triggered part of the *Rp*$^{+/-}$ responses, including increased p-eIF2α, and caused massive cell death (Figures 5A, S5A, and S5B)[18,24,25,31,46]. Xrp1^{long} activates an *Xrp1*-lacZ reporter [16,18], and overexpression of either isoform increased endogenous HA-tagged Xrp1, consistent with autoregulation (Figures S5C, S5D, control S1B). Both transgenes are driven by UAS/Gal4 motifs and include the hsp70 core promoter (UAS-hsp70), while contain only the Xrp1 coding sequences, lacking regulatory elements, such as the 5' and 3' UTRs that may mediate post-transcriptional control[47].

Twin-clone analysis showed that Xrp1^{long}-overexpressing clones were eliminated, whereas their sister clones survived[18]. Likewise, upon clonal overexpression of either isoform only tiny clones survived (Figures S5E and S5F, and Tsurui-Nishimura et al[46]). Even in an Xrp1-null background (using the *Xrp1*^{KO} deletion allele) and using a weaker act>y+>gal4 cassette, overexpression of either isoform led to clone elimination, with numerous basal apoptotic cells (Figures 5B-D). By contrast, GFP control clones were readily recovered without detectable cell death (Figures 5E). Under these conditions, it was not possible to determine whether Xrp1 induced death cell-autonomously or through competition.

Increasing the number of loser cells can enhance their survival in mosaics[48,49]. We therefore extended the heat-shock period from 25 to 45 minutes to generate more Xrp1-expressing cells (Figure 5F, see methodology). Under these conditions, Xrp1^{short} overexpression in either *Xrp1*$^{KO/KO}$ (Figure 5G-H, S5I-K, 7A-C) or *Xrp1*-intact (Figure S5G) discs produced mosaics in which the disc proper was largely populated by Xrp1^{short}-expressing cells. These cells were intermingled with wild-type neighbors and showed scattered cDcp1-positive cells, predominantly at clone boundaries (Figure 5G-I, S5G-K, 7A-C), both features of competitive interactions[9,10,50]. The basal layer displayed numerous apoptotic Xrp1^{short} losers, consistent with basal extrusion upon competition (Figure 5G and 5H). In some discs, the peripodial layer (squamous epithelium overlaying the disc proper) showed extensive apoptosis of Xrp1^{short} losers adjacent to wild-type neighbors (Figures 7B, 7C, S5G and S5H). This pattern highlights the context-dependent nature of cell competition, with elimination being stronger in the heterogeneous peripodial layer (Figure 7C and S5H) than in the relatively homogeneous disc proper (Figure 7B and S5G), where loser-loser interactions may provide protection.

Obtaining intermediate-sized Xrp1-overexpressing clones, where clones survived but did not cover the entire disc proper, proved challenging. Nevertheless, these discs revealed clear boundary-associated cell death (Figure 5I and 7A).

To further assess whether Xrp1 alone is sufficient to trigger cell competition, we used the *Xrp1*$^{Ex\text{-long}}$ deletion allele[47], hereafter referred to as *Xrp1*$^{UAS\text{-}short}$. This allele removes all *Xrp1*^{long} transcripts, but retains a UAS-containing enhancer upstream of the *Xrp1*-RD locus (Figure S3A)[47,51-53]. Homozygous *Xrp1*$^{UAS\text{-}short}$ discs showed modest background *Xrp1* activation even without Gal4 (Figure S6A). However, *RpS17*$^{+/-}$ mutants homozygous for *Xrp1*$^{UAS\text{-}short}$ did not upregulate Xrp1^{short} (Figure 3B). To minimize background expression, subsequent experiments used heterozygous *Xrp1*$^{UAS\text{-}short}$ wing discs or transheterozygotes with *Xrp1*^{KO} allele. Homozygous *Xrp1*$^{UAS\text{-}short}$ discs were used only in Figures S6O-S6P.

Xrp1 protein was strongly induced upon clonal Gal4 expression of *Xrp1*$^{UAS\text{-}short}$, confirming that the UAS elements can drive *Xrp1*-RD expression from its own promoter (Figure 5J). Clones were recovered even after brief heat shock, indicating that Xrp1 levels remained within a survival range (Figure 5J-L, S6D-S6L). In the wing pouch, *Xrp1*$^{UAS\text{-}short}$ induced Xrp1-HA expression (Figure S6B) and caused only mild cell death compared with the UAS-hsp70 Xrp1 transgenes (Figure S6C vs 5A and S5A). Notably, Xrp1 expression was reduced near the dorsoventral boundary (Figure 5J), resembling the Xrp1-HA pattern seen in *Rp*$^{+/-}$ cells (Figure S8B, also Figure 6 in Kiparaki et al[19]). Importantly, *Xrp1*$^{UAS\text{-}short}$-overexpressing clones showed pronounced boundary cell death, which was *Xrp1*-dependent (Figure 5K-M, S6D-S6L), demonstrating that Xrp1^{short} alone is sufficient to induce cell competition, in both wing and eye discs (Figure S6O-S6P).

Xrp1-overexpressing clones from either *UAS-hsp70 Xrp1* transgenes or *Xrp1*$^{UAS\text{-}short}$ were recovered only with a weaker *actin-FLP-out* cassette (the act>y+>Gal4) than that used in the RpS12 competition assays (act>CD2>Gal4) (see Methodology). Whether combining *Xrp1*^{KO} with the stronger *actin-FLP-out* cassette (act>CD2>Gal4) to disrupt autoregulation would enable recovery of Xrp1-overexpressing clones remains to be tested.

Xrp1^{long} overexpression in a homozygous *Xrp1*^{KO} background also triggered competition (Figures 5D). Clones were recovered only after prolonged heat-shock, and many discs contained Xrp1^{long} losers covering almost the entire disc proper, with few apoptotic cells mainly at clone boundaries (Figure S5L, 7D and S9G-I). By contrast, the peripodial layer of many discs showed numerous apoptotic losers adjacent to wild-type cells, resembling the pattern seen for Xrp1^{short} (Figure S5M). Thus, both Xrp1 isoforms can transform wild-type cells into losers, initiating cell competition.

Syncrip mediates fitness differences in *wild-type* and *Rp*$^{+/-}$ cells by regulating Xrp1^{short} expression.

RBPs equilibrium is finely regulated, and alterations in RBP abundance can profoundly impact RNA processing[54]. We hypothesized that *Rp*$^{+/-}$-specific changes in RBPs might depend on, or cooperate with, RpS12 in inducing Xrp1^{short}. Because Xrp1^{short} induction in *Rp*$^{+/-}$ cells occurs independently of Xrp1 activity we performed proteomic analysis of (a) *wild-type*, (b) *RpS3*$^{+/-}$, and (c) *Xrp1*$^{+/-}$, *RpS3*$^{+/-}$ wing discs (Figure S7). This analysis identified numerous differentially expressed RBPs, including Syncrip (Syp), previously shown to bind *Xrp1* mRNA[42]. Syp was downregulated in both *RpS3*$^{+/-}$ and *Xrp1*$^{+/-}$, *RpS3*$^{+/-}$ cells (Figures S7C-S7D).

We tested whether reduced Syp levels were responsible for Xrp1 activation in *Rp*$^{+/-}$ cells. Indeed, Syp reduction using two independent RNAi lines induced Xrp1 in wild-type discs, with one line producing a stronger induction (Figures 6A, 6C, control S8A). However, in *RpS18*$^{+/-}$ discs only the stronger line further increased Xrp1 (Figures 6B, 6D, control S8B), consistent with reduced endogenous Syp making *Rp*$^{+/-}$ cells less responsive to Syp depletion.

Syp knockdown conferred loser status on wild-type cells in an Xrp1-dependent manner(Figures 6E-6F, S8C and S8D). Indeed, Syp-depleted cells, similar to eIF6-depleted cells, increased the Xrp1^{short} levels (Figure 6G, S8D and S8E). eIF6-depleted cells are also Xrp1-dependent losers[19]. Syp depletion induced higher Xrp1 levels than single-copy RpS12 overexpression (Figure 6G and S8E). These results indicate that Syp restrains Xrp1^{short} expression, thereby influencing competitiveness of wild-type cells and likely contributing to the competitive disadvantage of *Rp*$^{+/-}$ cells (Figure 6H and S8F).

Xrp1 protein is post-transcriptionally degraded in surviving loser cells

In multiple discs, an apparent paradox emerged: when Xrp1 was overexpressed using the strong UAS-hsp70-driven transgenes and prolonged 45-min heat-shock, losers often survived in the disc proper and frequently occupied most of the tissue, with sporadic apoptosis (Figures 5G, S5G-S5M, 7B, 7D and S9). In the same discs, we could find loser cells in the peripodial layer presenting prominent boundary cell death next to wild-type neighbors (Figure S5H, S5M and 7C). The recovered clones were highly bimodal: either covering almost the entire disc proper (Figures 7B, 7D, S9D, and S9F) or were extremely small (similar to Figures 5B, 5D, S5E, and S5F), with intermediate-sized clones such as those in Figures 7A and 5I being rare.

To understand this outcome, we examined Xrp1 proteins in discs overexpressing Xrp1^{short} in an Xrp1^{KO} background. Unexpectedly, in discs where loser clones covered most of the disc proper, Xrp1 staining was often weak, patchy or even undetectable (Figures 7A, 7B, 7D, S9D, S9F and S9H). By contrast, cells in the peripodial epithelium of the same discs displayed strong Xrp1 expression, especially near cDcp1-positive cells at boundaries (Figure 7C and S9I). However, in the cells that did retain detectable Xrp1 staining, Xrp1 levels were much higher than in *Rp*$^{+/-}$ cells and *Xrp1*$^{UAS\text{-short}}$-expressing cells, as indicated by the need to lower confocal laser power to avoid signal saturation and also confirmed by immunoblotting (Figure 3A) .

In the few discs with intermediate-sized clones where boundary cell death could still be observed, Xrp1 was heterogeneous within clones and enriched at their boundaries (Figure 7A for the short and 7D for the long isoform). Importantly, the cells with the highest Xrp1 levels were not always the cDcp1-positive, indicating that elevated Xrp1 and apoptosis do not always coincide (Figure 7A-7D). Collectively, these findings suggest that surviving loser cells can adapt by post-transcriptionally reducing Xrp1 protein in a position-dependent manner. Cells in clone interiors, where loser-loser contacts predominate, reduce Xrp1, avoid autonomous elimination and survive, whereas boundary cells that contact wild-type neighbors maintain higher Xrp1 and undergo competitive apoptosis. This spatial pattern implies that the distance from winner cells may determine adaptive capacity.

DISCUSSION

RpS12-dependent splicing of Xrp1 initiates the *Rp*$^{+/-}$ phenotypes upstream of proteotoxic stress.

Our findings define the hierarchy among RpS12, Xrp1 and proteotoxic stress in the *Rp*$^{+/-}$ response[19,24,25,34,35]. RpS12-dependent induction of Xrp1^{short} is the initiating event and requires Gly97 for Xrp1 exon-4 skipping. Xrp1^{short}/Irbp18 autoregulation then increases all *Xrp1* transcripts and shapes the downstream *Rp*$^{+/-}$ responses (Figure 6H).

Although proteotoxic stress and p-eIF2α can activate Xrp1 in other contexts[19,24,25,29,34,35], Xrp1 is the driver of proteotoxic stress in *Rp*$^{+/-}$ cells and increases p-eIF2α, whereas its levels do not depend on p-eIF2α (Figure S3 and Kiparaki et al, 2022,[19]).

The most abundant RA/RG long transcripts produce little detectable protein, likely due to uORF-mediated translational regulation[28,29,44] (Figures S3C and S3D). In *Rp*$^{+/-}$ cells, RA/RG transcripts are transcriptionally autoregulated in an RpS12- and Xrp1-dependent manner. By contrast, the less abundant *Xrp1-RC* transcript is strongly induced in *Rp*$^{+/-}$ discs in an RpS12-dependent manner and produces Xrp1^{short} protein robustly (Figure 3). *Xrp1-RC* shares the RA/RG promoter but lacks the exon-4 uORFs.

Importantly, RpS12-dependent splicing occurs independently of Xrp1 protein, indicating that it does not require uORF-mediated translation or Xrp1 autoregulation.

p-eIF2α appears protective in *Rp*$^{+/-}$ cells[19,34,35], possibly by promoting low-level uORF-mediated translation of Xrp1^{long}, which may have some protective function. Supporting this, selective Xrp1^{long} depletion in *Drosophila* cells reduced growth and viability, whereas depletion of both isoforms did not [55]. Future studies will define the distinct roles of Xrp1 isoforms in *Rp*$^{+/-}$ responses and test whether uORF-dependent Xrp1^{long} translation occurs in other RpS12-independent contexts[19,26,32,56].Notably, in wild-type cells basal autoregulation depends only on Xrp1^{long}.

Ribosomopathies arise from impaired ribosome synthesis or function and commonly involve p53 activation and altered translation [57-61]. In *Drosophila*, Xrp1 mediates most *Rp*$^{+/-}$ phenotypes independently of p53, despite being a p53 target in DNA damage responses (DDR)[15,62-64]. *Rp*$^{+/-}$ cells accumulate ribosome biogenesis intermediates[19], which may trigger RpS12-dependent Xrp1 activation akin to the mammalian p53 nucleolar stress response[15,21,65,66].

RpS12–Xrp1^{short} splicing resembles RpL26-dependent splicing of the senescence-associated p53β isoform in mammalian DDR [67]. RpS12 also influences senescence in yeast[68]. Thus, ribosome-linked regulation of stress-responsive splice isoforms may be an evolutionarily conserved quality-control mechanism, with Xrp1^{short} in flies and p53β in mammals as analogous outputs. Whether similar RP-mediated splicing mechanisms regulate p53 in ribosomopathies merits future investigation.

Overexpression of RpS12 is sufficient to activate the Xrp1-dependent competition

Overexpression of RpS12 is sufficient to transform cells into “losers” even without *Rp* haploinsufficiency (Figure 1 and 4). The inability of other ribosomal proteins to induce Xrp1 (Figure 1) supports that RpS12 has a specialized role in loser specification, rather than causing a general perturbation of proteostasis. Thus, RpS12 is not merely permissive but instructive in promoting Xrp1^{short} production. In this view, *Minute* phenotypes arise not simply from loss of ribosomal function, but also from the gain of a specific RpS12 activity.

Moreover, RpS12 acts in a dose-dependent manner as one UAS-RpS12 induces modest Xrp1 expression, whereas two copies elicit enhanced Xrp1 induction and extensive cell competition (Figure 4). Notably, one UAS-RpS12 copy is sufficient to trigger competitive elimination, without detectable Xrp1 autoregulation or increased p-eIF2α. Thus, consistent with previous studies, neither Xrp1 autoregulation[26] nor increased p-eIF2α[19] is required for cell competition.

Xrp1 is sufficient to induce loser status.

We demonstrate that Xrp1 is not only necessary but also sufficient to induce “loser” status in wild-type cells. Thus, Xrp1 is the primary instigator of cell competition in $Rp^{+/-}$ cells, rather than simply a component of the death program. Overexpression of either isoform is sufficient to promote the loser identity in wild-type cells. However, the $\mathrm{Xrp1}^{short}$ isoform appears to be the functional one in $Rp^{+/-}$ cells. $\mathrm{Xrp1}^{long}$ may have a role in other Xrp1-dependent cell competition contexts, where RpS12 is not involved[19,26,32].

The proteomic data will be also instructive in distinguishing direct consequences of Rp haploinsufficiency from Xrp1-dependent responses and highlight candidate *Xrp1* splicing regulators, including Syncrip.

Syncrip is involved in Xrp1 regulation and cell competition

Syncrip (CG17838/Syp), the Drosophila homolog of mammalian SYNCRIP/hnRNPQ, is involved in RNA processing, transcript stability, and localized translation[69-71]. Syp is reduced in $Rp^{+/-}$ discs independently of Xrp1 and Syp depletion in otherwise wild-type cells is sufficient to induce $\mathrm{Xrp1}^{short}$-dependent cell competition (Figure 6). Notably, human SYNCRIP interacts with many RPs, including RpS12[72], as well as with the spliceosome[73]. Other RBPs, which bind the same Xrp1 intron as Syp[42], are also altered in $Rp^{+/-}$ cells (Figure S7). Therefore, reduced Syp may promote exon-4 skipping by permitting these factors to act (Figure S8F).

SYNCRIP depletion in human haematopoietic stem cells (HSC) reduces their competitive fitness and correlates with elevated CHOP[74], a factor with functional similarities to Xrp1 in stress and cell competition[7,16,18,25,29], suggesting a conserved mechanism where Syncrip and CHOP/Xrp1 factors regulate cellular fitness.

Moreover, Xrp1 mediates the toxicity of the RNA-binding protein Cabeza [47]. Given its large mRNA size and high RBP affinity[42], we speculate that Xrp1 may function as an “RBPostasis sensor”, detecting RBP imbalances and promoting elimination of affected cells. Consistently, Xrp1 also drives stress responses in cells with spliceosome defects[27].

Xrp1 levels are dynamically downregulated in a position-dependent manner

Our data reveal that Xrp1 protein levels are dynamically regulated depending on cellular position and clone size. Whereas cell competition has largely been viewed as a winner-driven elimination of losers, our findings suggest that loser status is influenced by adaptive responses to the local environment, including Xrp1 downregulation, which may reduce stress and avoid elimination.

When Xrp1 is overexpressed in mosaic epithelia, cells display three distinct behaviors: efficient elimination, boundary-restricted competition, or adaptation with clone survival. Clones expressing modest Xrp1 levels from the $\mathit{Xrp1}^{\mathit{UAS\text{-}short}}$ allele largely avoid autonomous apoptosis and are instead eliminated mainly through competitive interactions at clone boundaries, resembling the $\mathit{Rp}^{+/-}$ behavior (Figure 5 and S6). In contrast, strong Xrp1 expression driven by UAS-hsp70 transgenes either triggers widespread apoptosis when only few Xrp1-expressing cells are present or becomes actively downregulated once their initial number exceeds a threshold, thereby permitting cell survival (Figure 5 and S5).

A key feature of this adaptation is spatial Xrp1 heterogeneity. In large clones, interior disc-proper cells reduce Xrp1 to low or undetectable levels, whereas some boundary cells adjacent to wild-type neighbors retain high Xrp1 and boundary loser cell death occurs (Figures 7 and S9). In other cases, disc proper clones present heterogeneous or even undetectable Xrp1 levels, whereas same-genotype clones in the peripodial epithelium retain high Xrp1 and undergo apoptosis at interfaces with wild-type cells (Figure 7). In these clones, the cDcp1-positive cells do not coincide with the cells displaying the highest Xrp1 levels; instead, high Xrp1 is often detected in adjacent cells. Because both Xrp1 and GFP transgenes are transcribed under Gal4-UAS control, this mismatch indicates that Xrp1 is subject to post-transcriptional downregulation. The initial abundance of Xrp1-overexpressing cells, their position and their interactions with neighboring cells then determine whether cells die or survive.

Xrp1 downregulation upon stronger Xrp1 induction raises an important mechanistic question for future investigation. One plausible model is that Xrp1 activates adaptive responses that alleviate cellular stresses and limit its own accumulation. We speculate that this could occur through negative feedback mechanisms involving Xrp1 degradation, similar to p53-MDM2 feedback regulation[75]. Combined with Xrp1/Irbp18 autoregulation and other post-transcriptional mechanisms, such feedback regulation may keep the circuit active but yet self-limiting. Consistent with this, Xrp1 induction from UAS-hsp70 transgenes, which lack endogenous transcriptional and post-transcriptional regulatory elements, results in predominant downregulation. In contrast, the $Xrp1^{UAS\text{-}short}$ allele and $Rp^{+/-}$ contexts, which retain cis-regulatory elements, maintain a more balanced interplay between induction and downregulation, permitting survival under lower Xrp1 expression levels(Figure 7E).

We propose that adaptive downregulation resembles a quorum-like process, in which number of loser cells and the identity of their neighbors determine whether the loser program is sustained or suppressed. In this context the signals are likely short-range rather than diffusible autoinducers as in classical bacteria quorum sensing [76,77] or other quorum-like systems in multicellular organisms[78,79]. In this framework winners could act as quorum-quenchers, suppressing adaptive stress responses in neighboring loser cells. Consistent with this idea, L-proline may act as a quorum-quencher secreted by winner cells to eliminate losers with mitochondrial dysfunction by suppressing their cytoprotective ATF4/CHOP-mediated stress response pathway [80]. Such quorum-like response may indeed underlie the survival of losers in homogeneous environments.

Therefore, in addition to RpS12-dependent induction and Syncrip-mediated control of the $Xrp1^{short}$ isoform, dynamic Xrp1 downregulation emerges as a key determinant of whether cells die, compete, or adapt. This outcome is strongly influenced by the local environment and Xrp1 expression levels.

Human ribosomopathies combine tissue-specific defects with cancer predisposition[1,57,58,81-83]. Understanding how the $Rp^{+/-}$-dependent $Xrp1^{short}$-driven loser program is initiated, amplified, and eventually attenuated may illuminate how ribosome-linked stress responses contribute to early quality control and, when dysregulated, to clonal evolution and tumorigenesis.

Limitations of the study

Further research is needed to determine whether RpS12 regulates splicing directly as a *bona fide* splicing factor, like some homologs[84] or indirectly through translational control of splicing mediators(e.g.Syp)[85-87]. The initial Syncrip data presented herein leave open the possibility that RpS12 regulates *Xrp1* splicing through different mechanism. It will be important testing whether RpS12 downregulates Syncrip in $Rp^{+/-}$ cells and whether Syncrip controls Xrp1 through splicing or another post-transcriptional mechanism (Figure 6H). RpL10Ab homologs have been implicated in alternative splicing and specialized translation [85,88], raising the possibility that RpS12, RpL10A and Syncrip act within a broader Xrp1 regulatory program during cell competition. Although differences in 5′ UTRs could affect isoform stability[89], the most parsimonious explanation of our genetic and splicing data is RpS12-dependent *Xrp1* splicing rather than selective stabilization of pre-existing short transcripts.

RESOURCE AVAILABILITY

Lead Contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Marianthi Kiparaki (kiparaki@fleming.gr).

Materials Availability

This study did not generate new unique reagents.

Data and Code availability

- This paper reanalyzes publicly available RNA-seq data deposited in the Gene Expression Omnibus under accession numbers: GSE112864 and GSE124924 (see key resources table).
- The mass spectrometry proteomics data of this study have been deposited to the ProteomeXchange Consortium via the PRIDE[90] partner repository with the dataset identifier PXD076715.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

ACKNOWLEDGMENTS

We specially thank Nicholas Baker and Christos Delidakis for valuable comments on the manuscript. We also thank Chrysoula Pitsouli and Nektarios Tavernarakis for feedback on an earlier draft, and Maria Lalioti and Konstantina Dimakakou for technical support. We thank Tatsushi Igaki for sharing information prior to submission. We would like also to thank anonymous reviewers for their comments. We thank Christos Delidakis, Isabel Guerrero and Erik Storkebaum for providing flystocks, and Don Rio for the rabbit anti-Xrp1 antibody used in preliminary experiments. We are grateful to Hyung Don Ryoo for generously providing the guinea pig anti-Xrp1 antibody during the revisions. This study was supported by the Hellenic Foundation for Research and Innovation (HFRI) under the “3rd Call for HFRI Research Projects to support Post-Doctoral Researchers” to M.K.(Project Number: 7306). Salary support for M.P. by the HFRI (Fellowship:11141). Salary support for K.D., M.L., and E.M.C.S. by the HFRI (Project Number: 16975). This study was additionally supported by family funds of M.K. Confocal microscopy was performed using the Leica TCS SP8X WLL confocal system funded by the BIOIMAGING-GR (MIS 5002755) and the Zeiss LSM 900 Airyscan funded by the Regional Operational Programme 'ATTICA' (NSRF 2021-2027), ERDF, Greece/EU (MIS 6004752). Proteomic instrumentation supported by the Operational Programme “Competitiveness, Entrepreneurship and Innovation”, NSRF 2014-2020,ERDF, Greece/EU (MIS 5002802 pMedGR). Drosophila stocks were obtained from the Bloomington Drosophila

Stock Center (supported by NIH P40OD018537), the Vienna Drosophila RNAi Center[91] and FlyORF library[92].

AUTHOR CONTRIBUTIONS

E.T. and K.Kont. performed genetic and molecular experiments and analyzed the data. M.P. performed alternative splicing, isoform and RT-PCR/qPCR analyses. K.D. performed additional RT-PCR/qPCR experiments. N.V. and P.M. quantified the *Xpr1*$^{M2\text{-}73}$ allelic mutation. K.Kan. and M.L. contributed in fly experiments. E.M.C.S provided intellectual input. M.S. performed the proteomics experiments and analyzed the data. P.K. supervised the alternative splicing, isoform, and RT-PCR/qPCR experiments, contributed to data analysis, and edited the manuscript. M.K. conceptualized and led the study, designed and performed experiments, analyzed and interpreted all data, acquired funding, and wrote the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work, the authors used Perplexity AI in order to perform language editing. After using this tool or service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

FIGURE TITLES AND LEGENDS

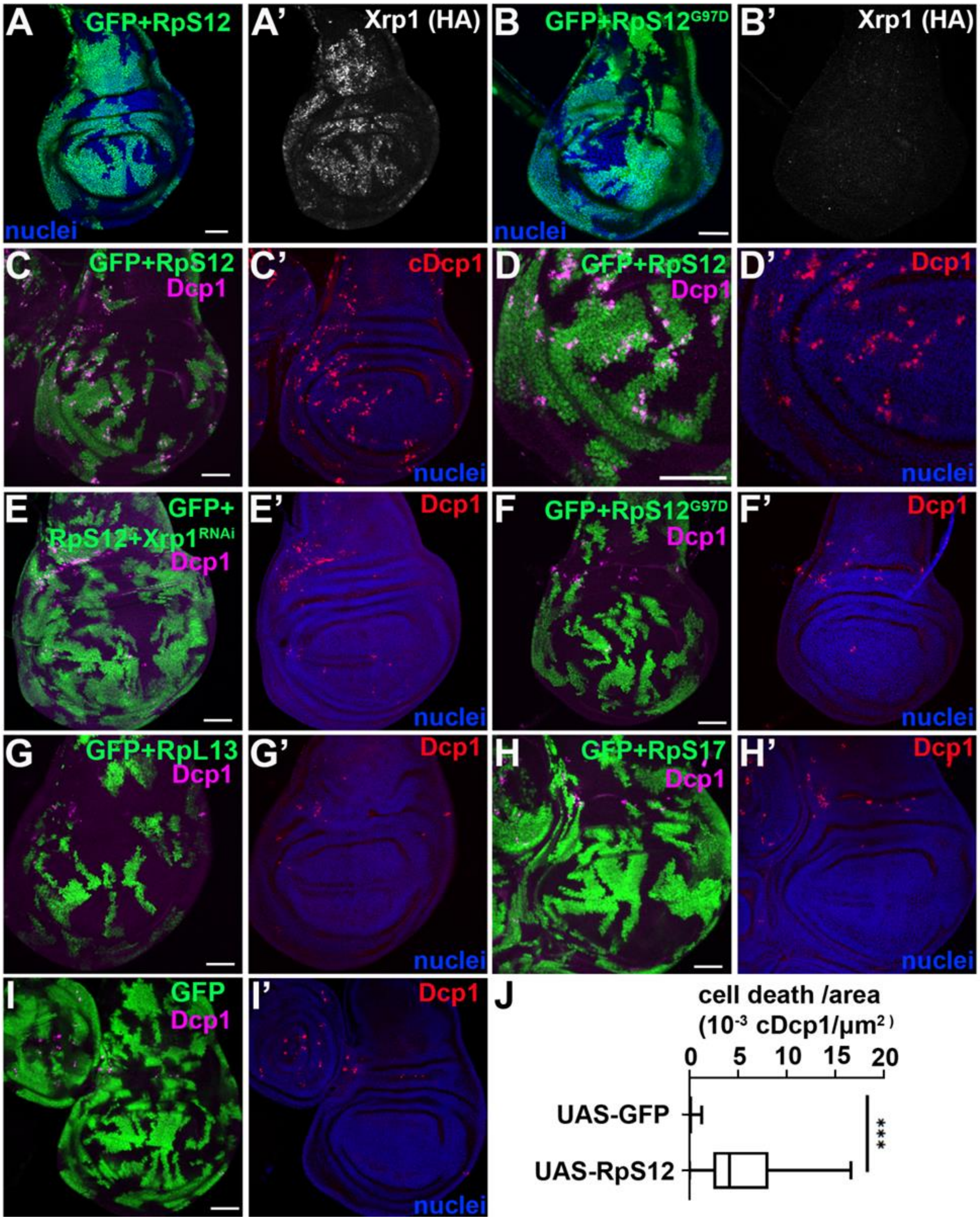


Figure 1 RpS12 overexpression is sufficient to induce Xrp1-dependent cell competition in wild-type cells.

Confocal images of third-instar wing discs containing clones generated with a FLP-out Gal4 and the indicated transgenes. A and B shows single confocal planes stained with anti-HA. C-I show projections of active cDcp1 staining in the central disc proper, shown in magenta or red. Clones are marked by UAS-GFP in green and nuclei are in blue. Scale bars, 50μm.

(A, B) RpS12-overexpressing clones (n=12) (A) induce Xrp1-HA expression (A'), whereas $RpS12^{G97D}$-overexpresing clones (n=6)(B) do not.

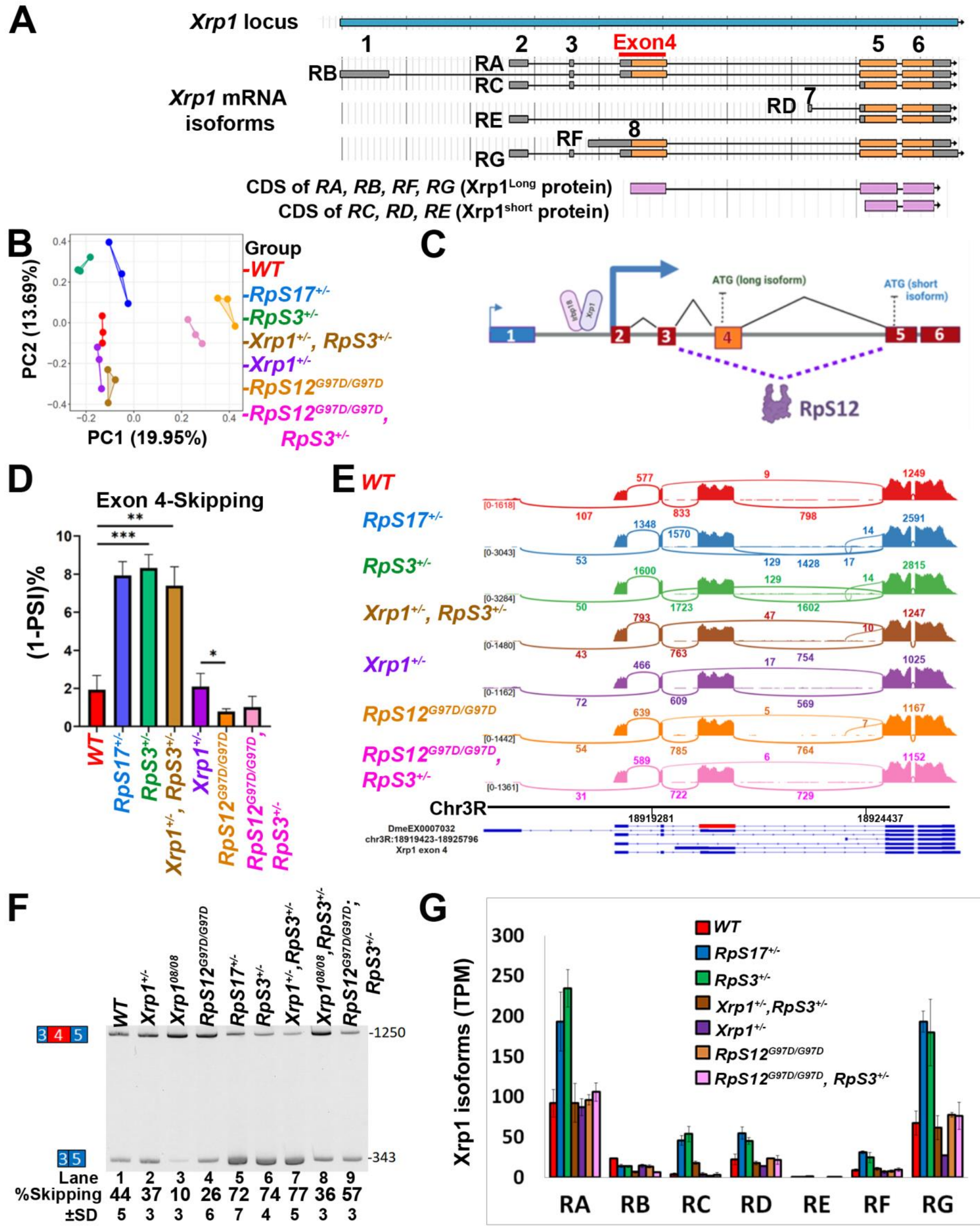
(C, D) RpS12-overexpressing clones induce boundary cell death (cDcp1) at interfaces with wild-type cells (n=10)(higher magnification in D).

(E) Clones co-expressing RpS12 and Xrp1-RNAi lack boundary cell death (n=8) (Dcp1) .

(F-I) Clones overexpressing of $RpS12^{G97D}$ (n=10)(F), RpL13 (n=5) (G), RpS17 (n=5) (H) or GFP alone (n=9) (I) in wild-type discs does not increase cell death.

(J) Cell death per area in RpS12-overexpressing and control GFP clones. Box-and-whisker plots show median, interquartile range, and minimum–maximum values for cell death/area for act>CD2>Gal4, UAS-GFP control and act>CD2>Gal4, UAS-GFP × UAS-RpS12 discs. Statistical significance is shown as follows: ***$p < 0.001$; Wilcoxon rank sum test.

See also Figure S1



493

494 **Figure 2. RpS12 promotes exon 4 skipping of *Xrp1* mRNA, producing the Xrp1^{short}.**

(A) The *Xrp1* locus has seven annotated transcripts.

(B) Principal Component Analysis of PSI values for all exonic splicing events across the indicated genotypes (PC1, 19.95 % variance and PC2 13.69% variance). Each point represents one biological sample. 3 biological samples analyzed for each genotype. Genotypes are color-coded as follows: wild-type (WT, red), *RpS17*$^{+/-}$ (blue), *RpS3*$^{+/-}$ (green), *Xrp1*$^{+/-}$*, RpS3*$^{+/-}$ (brown), *Xrp1*$^{+/-}$ (purple), *RpS12*$^{G97D/G97D}$ (orange), and *RpS12*$^{G97D/G97D}$*,RpS3*$^{+/-}$(pink).

(C) Model of RpS12-mediated *Xrp1* splicing. Exon 4 skipping generates the *Xrp1-RC* transcript, which encodes the Xrp1^{short}. In *Rp*$^{+/-}$ cells, Xrp1^{short}/Irbp18 heterodimer promotes autoregulation. Exon-4 contains the Xrp1^{long} translation start site (ATG), whereas the Xrp1^{short} translation initiates within exon 5.

(D) *Xrp1* exon 4 skipping values in the final mRNA, calculated as 1-PSI, shown as mean±SEM. *p-values* were calculated by Student's t-test (**p< 0.01, ***p< 0.001). Genotypes and color coding are as in B. 3 biological samples for each genotype.

(E) Sashimi plots of *Xrp1* alternative splicing events from aligned RNA-seq data. Arcs represent splice junctions, and labels indicate unique reads mapped to each junction. Junction reads reflect both splicing and transcriptional changes, consistent with *Xrp1* transcriptional autoregulation in *Rp*$^{+/-}$ cells. A schematic below depicts the exons-intron structure of the different *Xrp1* transcripts, with the chromosomal location indicated underneath. The red box marks exon-4 (Chr3R: 18919423-18925796; VastDB: DmeEX0007032). Genotypes and color coding are as in B. 3 biological samples were included for each genotype.

(F) Semi-quantitative RT-PCR analysis of Xrp1 exon 4 splicing in the indicated genotypes. Primers in exons 3 and 5 amplify products that either include (1250bp) or skip (343bp) exon-4. Skipping values represent the skipping band relative to the sum of inclusion and skipping bands. Error bars show ±SD from 3 biological replicates. Lanes: (1)*wild-type (WT)*, (2)*Xrp1*$^{M2-73/+}$, (3)*Xrp1*$^{08/08}$, (4)*RpS12*$^{G97D/G97D}$, (5)*RpS17*$^{+/-}$, (6)*RpS3*$^{+/-}$, (7)*Xrp1*$^{M2-73/+}$*,RpS3*$^{+/-}$, (8)*Xrp1*$^{08/08}$*,RpS3*$^{+/-}$, (9)*RpS12*$^{G97D/G97D}$*,RpS3*$^{+/-}$. For each genotype n=3.

(G) Isoform analysis of *Xrp1* transcripts from the RNA-seq datasets as in B, D, and E. The graph shows transcripts per million (TPM) values for each of the seven isoforms in the indicated genotypes. Error bars represent ±1 SD. 3 biological samples analyzed for each genotype.

See also Figure S2

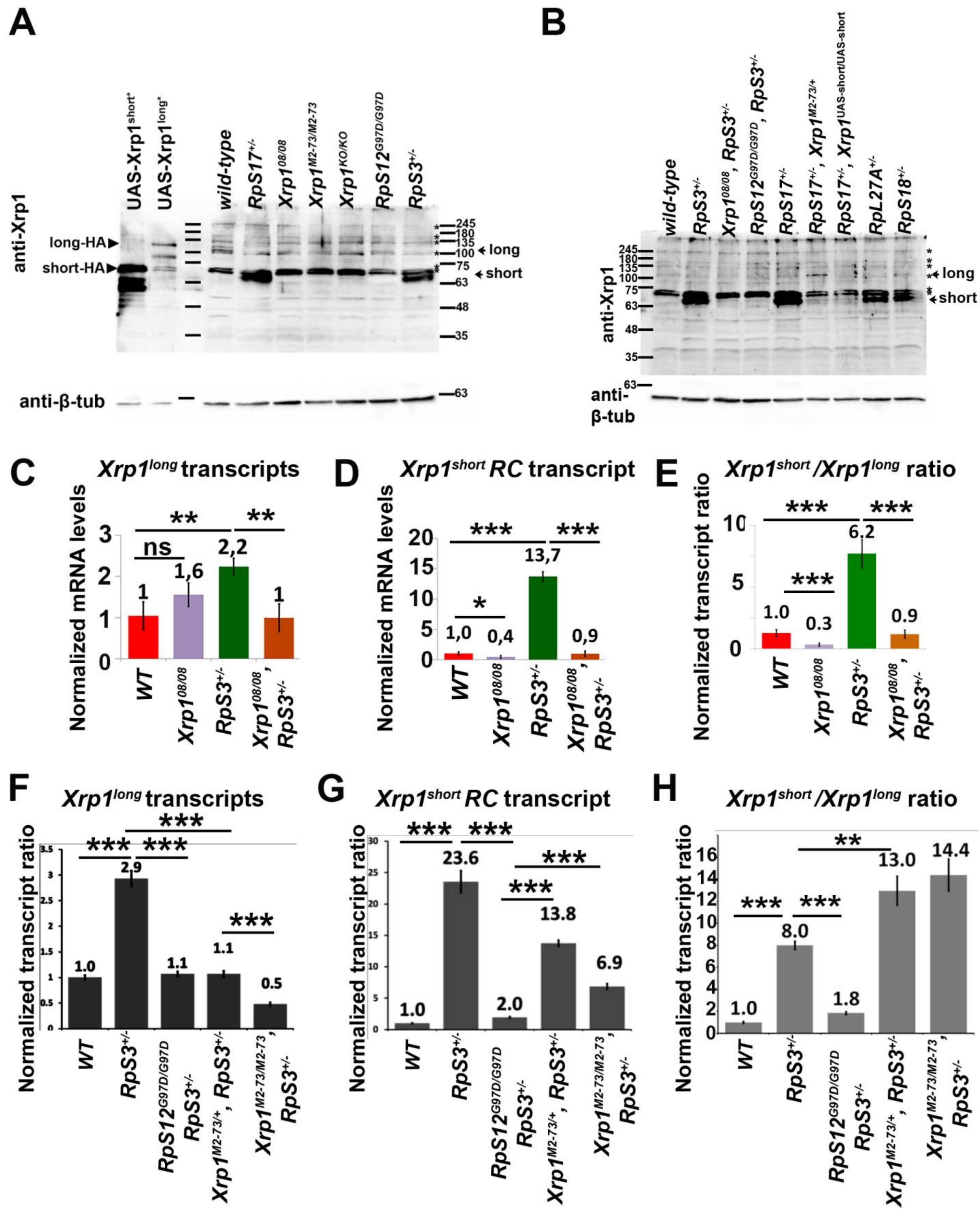


Figure 3. RpS12 promotes exon 4 skipping to induce Xrp1^{short}, which is required for cell competition.

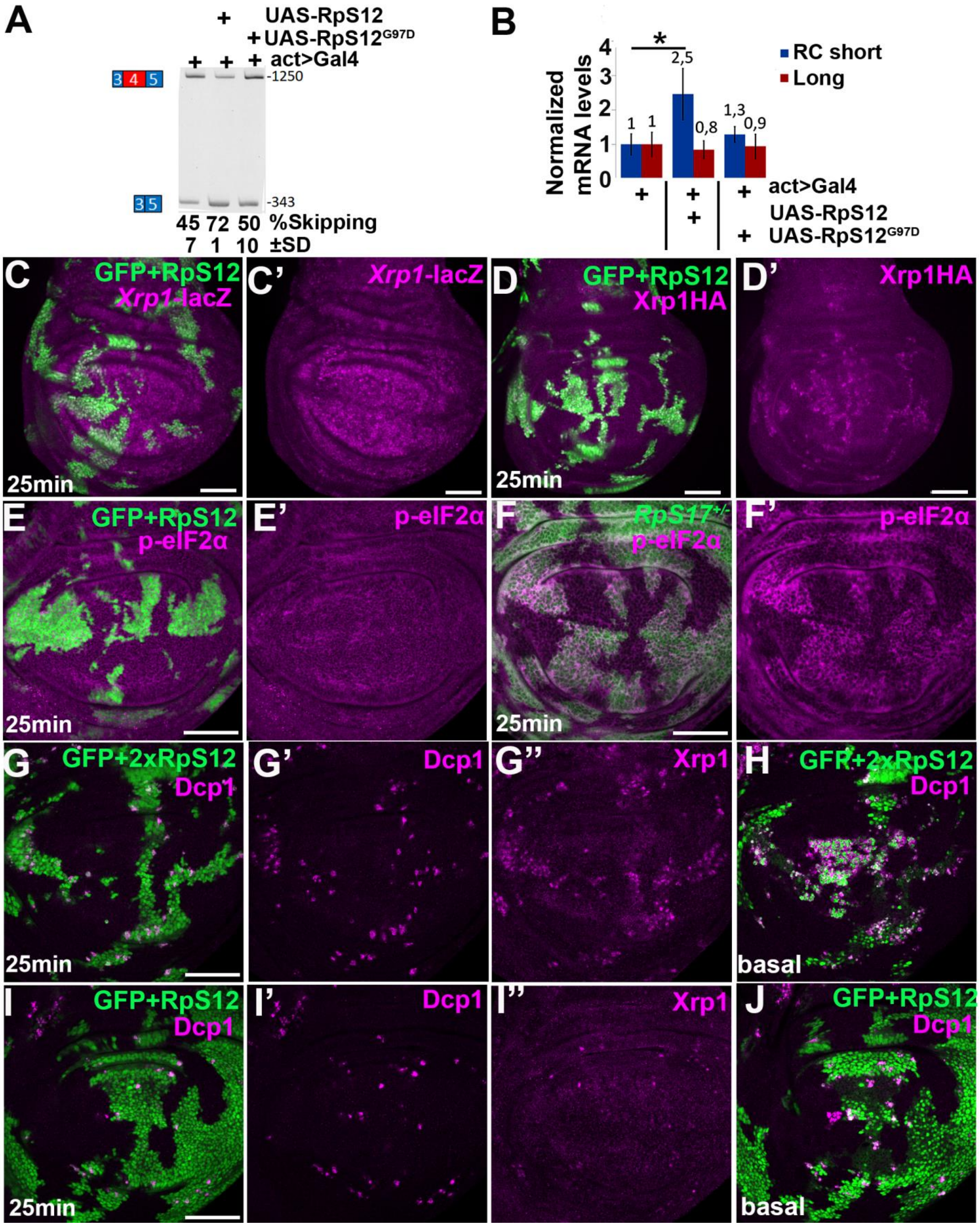
(A, B) Western blot analysis of Xrp1 protein in third-instar wing disc lysates from the indicated genotypes using anti-Xrp1 from Ryoo's lab. β-tubulin was used as a loading control. For each genotype, n=3, except $Xrp1^{UAS\text{-}short/UAS\text{-}short}$, $RpS17^{+/-}$ (panel B, lane 8) where n=2.

Lanes of panel (A): (1) UAS-Xrp1^{short}-HA (AT-hook mutant), (2) UAS-Xrp1^{long}-HA clones (AT-hook mutant), (3) protein ladder, (4) *wild-type*, (5) $RpS17^{+/-}$, (6) $Xrp1^{08/08}$, (7) $Xrp1^{M2\text{-}73/M2\text{-}73}$, (8) $Xrp1^{KO/KO}$, (9) $RpS12^{G97D/G97D}$, (10) $RpS3^{+/-}$. UAS-Xrp1-HA proteins migrate more slowly than endogenous Xrp1, because of the HA epitope.

Lanes of panel (B): (1) protein ladder, (2) *wild-type,* (3) $RpS3^{+/-}$, (4) $Xrp1^{08/08}$, $RpS3^{+/-}$, (5) $RpS12^{G97D/G97D}$, $RpS3^{+/-}$, (6) $RpS17^{+/-}$, (7) $Xrp1^{M2\text{-}73/+}$, $RpS17^{+/-}$, (8) $Xrp1^{UAS\text{-}short/UAS\text{-}short}$, $RpS17^{+/-}$, (9) $RpL27A^{+/-}$, (10) $RpS18^{+/-}$.

(C-H) qRT-PCR analysis of $Xrp1^{long}$ (C, F) and Xrp1^{short}-RC (D,G) transcripts, and the Xrp1^{short}-RC/Xrp1^{long} ratio (E,H), in the indicated genotypes. The forward primer for Xrp1^{long} spanned the exon4-exon5 junction; the forward primer for Xrp1^{short}-RC spanned the exon 3-exon 5 junction; for both transcripts the reverse primer annealed within exon 5. Error bars indicate ±1 SD from 3 biological samples, each with 2-3 technical replicates. p-values were calculated using Student's t-test (* $p< 0.05$, ** $p< 0.01$, *** $p< 0.001$). In C-E, the genotypes are: (1) *wild-type,* (2) $Xrp1^{08/08}$, (3) $RpS3^{+/-}$ and (4) $Xrp1^{08/08}$, $RpS3^{+/-}$. In F-H, the genotypes are: (1) *wild-type,* (2) $RpS3^{+/-}$, (3) $RpS12^{G97D/G97D}$, $RpS3^{+/-}$, (4) $Xrp1^{M2\text{-}73/+}$, $RpS3^{+/-}$, and (5) $Xrp1^{M2\text{-}73/ M2\text{-}73}$, $RpS3^{+/-}$.

See also Figure S3



549 **Figure 4 RpS12 overexpression is sufficient to induce Xrp1^{short} in wild-type cell in a dosage-**
550 **dependent manner**

(A) Semi-quantitative RT-PCR analysis of Xrp1 exon 4 in the indicated genotypes. Lanes: (1) *act>Gal4, UAS-GFP*, (2) *act>Gal4, UAS-GFP, UAS-RpS12*, (3) *act>Gal4, UAS-GFP, UAS-RpS12*^{G97D}. Conditions are the same as in Figure 2F. 3 biological samples were included for each genotype.
(B) qRT-PCR analysis of *Xrp1-RC* short (blue) and *Xrp1*^{long} (red) transcripts in wing discs overexpressing UAS-GFP alone or together with UAS-RpS12 or UAS-RpS12^{G97D} under act-Gal4. Primers are as in Figure 3C-3H. Error bars represent ±1 SD. p-values are calculated by Student's t-test (* $p< 0.05$). 3 biological samples for each genotype, with 2-3 technical replicas.

(C-J) Confocal images of third-instar wing discs. In panel F, wild-type clones were induced in an *RpS17*$^{+/-}$ background (labeled with anti-βgal in green) by mitotic recombination. In all other panels, discs contain clones generated with FLP-out Gal4, marked with UAS-GFP, and overexpressing UAS-RpS12. Single confocal plane are shown in the disc proper (C-G, I) or basal side (H,J). Scale bars, 50μm.

(C,D) RpS12-overexpressing clones do not induce the *Xrp1*-lacZ transcriptional reporter (n=5) (C, anti-βgal staining in magenta), but do induce Xrp1-HA expression (n=5) (D, anti-HA staining in magenta).

(E, F) RpS12-overexpressing clones do not elevate p-eIF2α (n=4)(E), whereas control *RpS17*$^{+/-}$ clones do (n=7)(F). Discs were stained with anti-p-eIF2α (magenta).

(G-J) Clones over-expressing two-copies of UAS-RpS12 induce stronger Xrp1 upregulation and more boundary cell death (n=5)(G) than clones expressing one copy of UAS-RpS12 (n=4)(I). Extensive apoptotic cell accumulation is observed on the basal side of 2xRpS12 discs (H), compared with RpS12 discs (I).Cell death labelled with active cDcp1 staining and Xrp1 was detected with anti-GP-Xrp1.

See also Figure S4

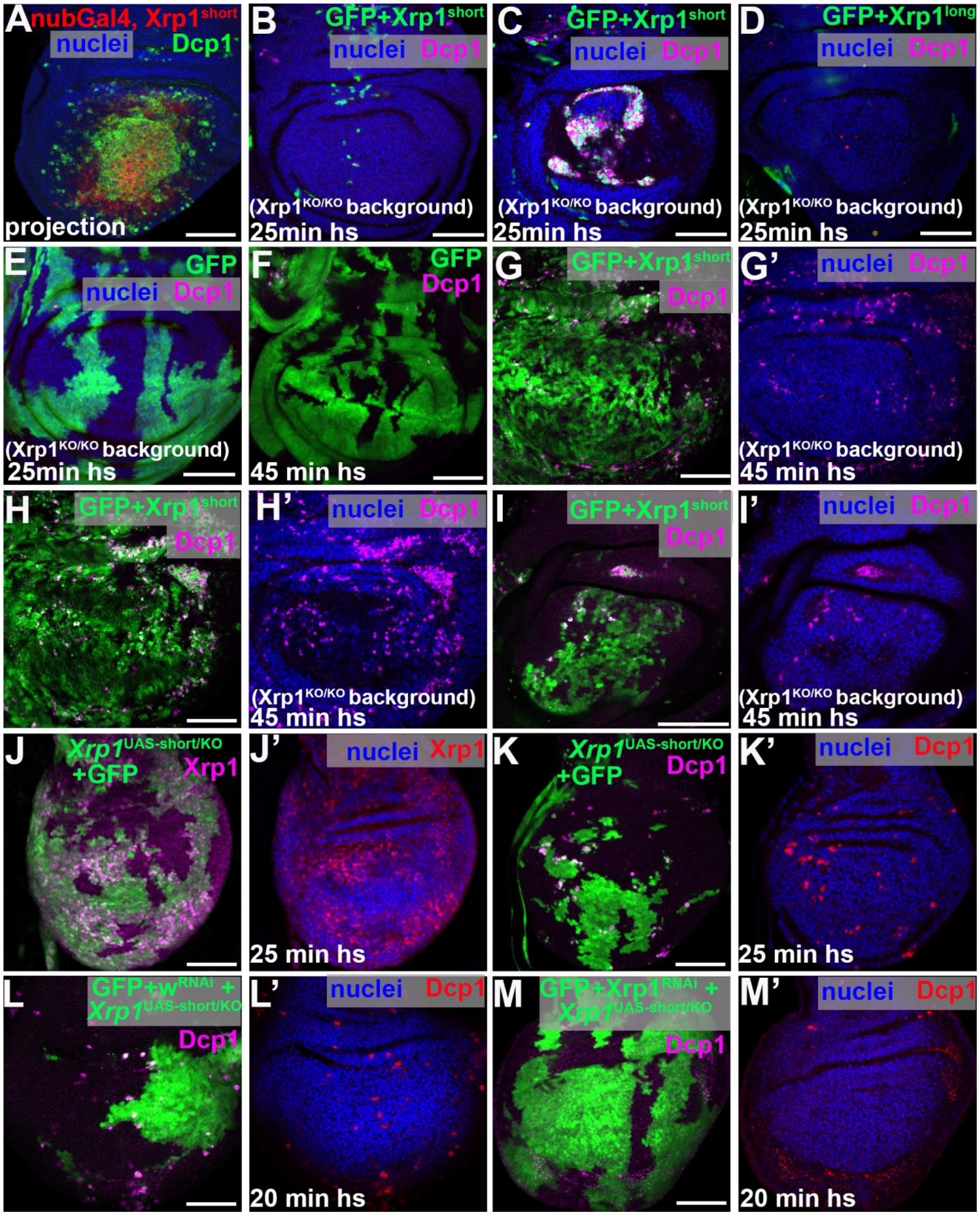


Figure 5 Overexpression of either Xrp1 isoform is sufficient to transform wild-type cells into losers

Confocal images of third-instar wing discs. Panels A-D show projections of the disc proper, except A, which also includes the basal side, and C, which shows only the basal side. Panels B-M shows discs containing clones generated with the weak FLP-out Gal4 cassette, marked with UAS-GFP (green), and overexpressing the indicated transgenes. Nuclei are shown in blue. Panels E-M show single confocal planes. Scale bars, 50µm

(A) nub-Gal4 drives expression of RFP (red) and Xrp1^{short} in the wing pouch, causing extensive cDcp1-positive cell death (green) (n=4).

(B, C) Xrp1^{short}-overexpressing clones in *Xrp1*$^{KO/KO}$ wing discs do not survive (B) and show massive basal cell death (C), detected by cDcp1. Clones were induced by a 25-min heat shock (n=6).

(D, E) Xrp1^{long}-overexpressing clones in *Xrp1*$^{KO/KO}$ wing discs are eliminated (n=5) (D) (cDcp1, magenta), whereas control clones do not show cDcp1-positive cells (n=8)(E). Clones were induced by a 25-min heat shock.

(F) Control clones overexpressing GFP do not show show cDcp1-positive cells (n=9). Clones were induced by a 45-min heat shock.

(G, H) After a 45-min heat-shock, Xrp1^{short}-overexpressing clones in *Xrp1*$^{KO/KO}$ wing discs can survive and show boundary cell death (n=14) (G) (cDcp1, magenta). On the basal side (H), more cDcp1-positive cells accumulate in the posterior compartment, where loser−wild-type interfaces are more than in the anterior.

(I) An intermediate-sized Xrp1^{short}-overexpressing clone in Xrp1$^{KO/KO}$ wing discs shows prominent boundary cell death (cDcp1, magenta) (n=3). Clones were induced by a 45-min heat shock.

(J, K) Clonal overexpression of Xrp1 from the *Xrp1*$^{UAS\text{-}short}$ allele in *Xrp1*$^{UAS\text{-}short/KO}$ wing discs leads to Xrp1 upregulation (n=7)(J) and boundary cell death (n=5) (K). Clones were induced by a 25-min heat shock.

(L, M) In *Xrp1*$^{UAS\text{-}short/+}$ wing discs, Xrp1 overexpression in clones induces boundary cell death (n=5) (L), which is abolished by co-expression of Xrp1-RNAi (n=4)(M). Clones were induced by 20-min heat shock.

See also Figure S5 and S6

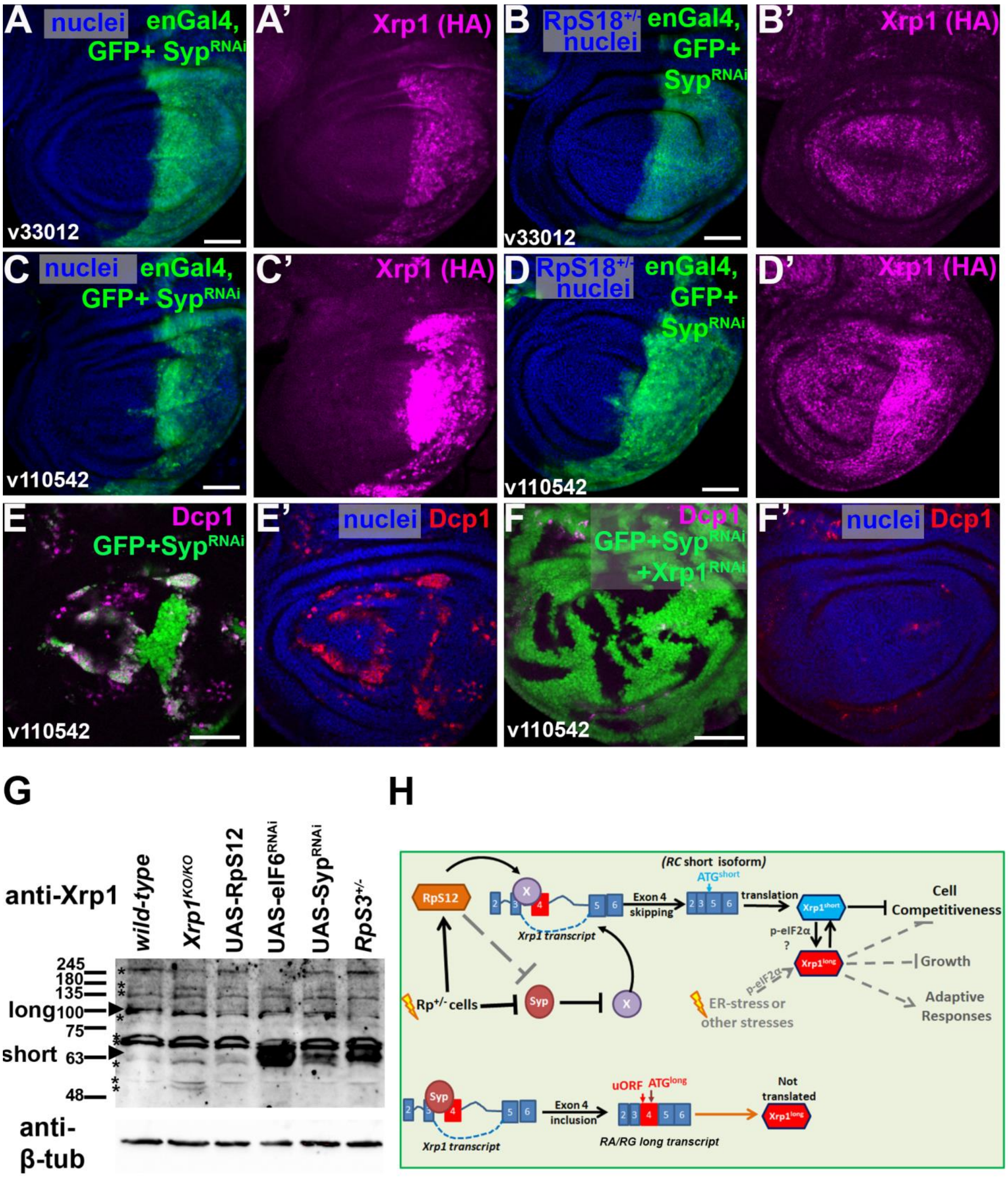


Figure 6. Syncrip regulates Xrp1 expression and cell competition

Confocal images of wing discs. Panels E and F show projections of cDcp1 staining in the disc proper. All other panels show single confocal planes. Scale bars, 50µm.

(A-D) en-Gal4 drives coexpression of GFP (green) in the posterior domain of wild-type (A,C) or *RpS18*$^{+/-}$ (B,D) wing discs together with Syp-RNAi. The SypRNAi v33012 line induces Xrp1-HA expression in wild-type cells (n=6)(A), but does not further increase Xrp1-HA levels in *RpS18*$^{+/-}$ cells (n=6)(B). The stronger Syp-RNAi v110542 line induces higher Xrp1-HA expression in wild-type cells (n=5) (C) than v33012 and also induces Xrp1-HA in *RpS18*$^{+/-}$ cells (n=6)(D).

(E) Clones coexpressing Syp-RNAi (v110542) and GFP in wild-type discs show boundary cell death detected by cDcp1 (n=6) (magenta in E and red in E'). Clones were induced by a 45-min heat-shock. No clones were recovered after a 25-min heat-shock (not shown).

(F) Boundary cell death in clones expressing Syp-RNAi (v110542) and GFP, is completely abolished by coexpression of Xrp1-RNAi (n=6)(cDcp1, magenta in F and red in F').

(G) Western blot analysis of Xrp1 protein (using anti-Xrp1 from Ryoo's lab) in third-instar wing disc lysates from the indicated genotypes. Lanes: (1) *wild-type discs*, (2) Xrp1$^{KO/KO}$ discs, (3) *actGal4, UAS-RpS12 discs* (4) *discs with act>CD2>Gal4, UAS-eIF6-RNAi* clones (5) *discs with act>CD2>Gal4, UAS-Syp-RNAi* clones, and (6) *RpS3*$^{+/-}$ *discs*. The Xrp1^{short} isoform is strongly induced in eIF6-depleted cells, in Syp-depleted cells, and *RpS3*$^{+/-}$ discs. The Xrp1^{long} isoform is barely detectable in wild-type cells and in discs containing eIF6- or Syp-depleted clones. The band just below Xrp1^{long} in Xrp1$^{KO/KO}$ null lane represents background. Other background bands are marked with asterisks. β-tubulin was used as a loading control.(n=3, except samples eIF6-RNAi=2 and Syp-RNAi=2). The whole blot is shown in Figure S8.

(H) Proposed model involving Syncrip: In *Rp*$^{+/-}$ cells, RpS12 promotes exon 4 skipping, generating the *Xrp1-RC* transcript, which lacks uORF and is efficiently translated into the Xrp1^{short} isoform. Xrp1^{short} is required to convert *Rp*$^{+/-}$ cells into losers. In *Rp*$^{+/-}$ cells, Xrp1^{short} protein increases the levels of all Xrp1 transcripts. The abundant *Xrp1-RA/RG* isoforms remain largely untranslated, likely due to uORFs in their 5' UTR. Syncrip is proposed to inhibit exon-4 skipping in *wild-type* cells, and its reduced levels in *Rp*$^{+/-}$ cells may contribute to Xrp1^{short} production. Syp may also repress Xrp1 through a splicing-independent mechanism. RpS12 may mediate Syp reduction in *Rp*$^{+/-}$ cells. Either Xrp1 isoform is sufficient to induce cell competition when expressed in cells. Other stresses, such as endoplasmic reticulum (ER) stress, via increased p-eIF2α, may enable *RA*/*RG* translation and subsequent induce Xrp1^{short} through autoregulation.

See also Figure S7 and S8

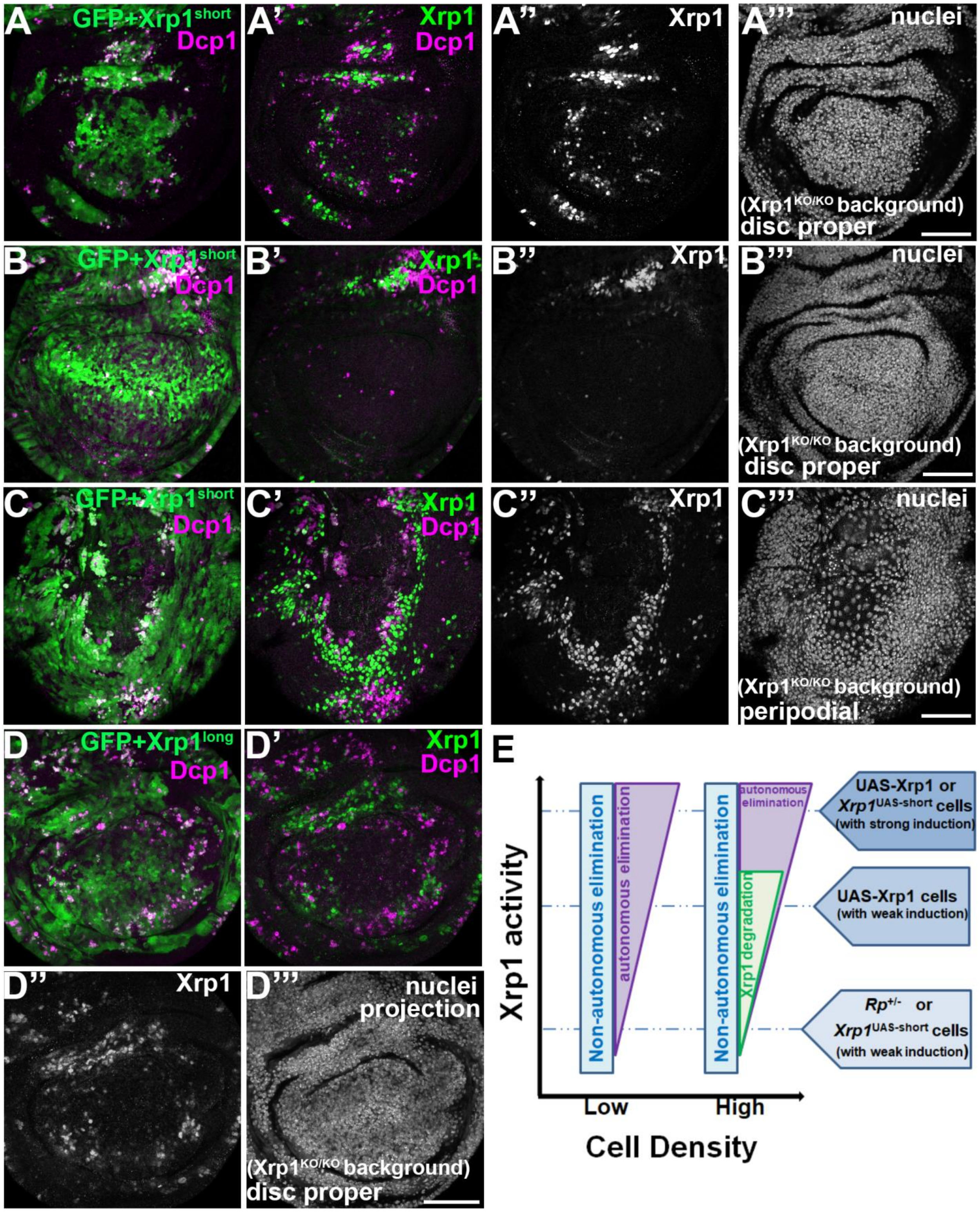


Figure 7 Xrp1 protein is post-transcriptionally degraded in surviving loser cells.

Confocal images of third-instar wing discs. Panels A-C show single sections, whereas panel D shows a projection. Clones were induced by a 45 min heat-shock and generated with the weak FLP-out Gal4 cassette. Clones expressed Xrp1 transgenes together with UAS-GFP (green in A-D). Nuclei are shown in blue. Panel E-M shows single confocal planes. n=14 for Xrp1^{short} samples and n=10 for Xrp1^{long} samples. Scale bars, 50μm

(A, D) Xrp1^{short} or Xrp1^{long} overexpressing clones in Xrp1$^{KO/KO}$ wing discs show cDcp1-positive cell death (magenta) at clone boundaries adjacent to wild-type cells. Xrp1 expression is nonuniform (green in A', D' and white in A'', D''), with elevated signal near cDcp1-positive boundary cells (magenta) and reduced signal in clone centers. Clones were analyzed 3 days after induction.

(B, C) In a representative disc with 2-day induced clones, Xrp1^{short} overexpressing cells in the disc proper show barely detectable Xrp1 and cDcp1 signals in the wing pouch (B' and B''). In the peripodial membrane of the same disc, Xrp1^{short} overexpressing cells undergo boundary cell death when adjacent to wild-type cells (C), with Xrp1 expression increased near cDcp1-positive cells (C' and C''). Xrp1 levels are also nonuniform in the peripodial clones.

(E) Schematic representation of Xrp1-dependent outcomes, distinguishing cell competition from autonomous cell death depending on the Xrp1 activity and loser-cell numbers (density). At low loser-cell density, low Xrp1 activity promotes mainly cell death through cell competition, whereas higher Xrp1 levels trigger autonomous elimination. At high loser-cell density, low Xrp1 activity predominantly drives non-autonomous elimination, as observed in *Rp*$^{+/-}$ and Xrp1$^{UAS\text{-}short}$-overexpressing cells. In contrast, at higher loser density, a quorum-like behavior protects cells with increased Xrp1 induction from autonomous elimination through post-transcriptional downregulation of Xrp1. However, elimination becomes unavoidable once Xrp1 levels exceed a critical threshold.

See also Figure S9

STAR★METHODS

KEY RESOURCES TABLE

REAGENT OR RESOURCE	SOURCE	IDENTIFIER
Antibodies		
anti-Xrp1 (guinea-pig polyclonal)	Ryoo lab generated guinea pig polyclonal	[29]
anti-Xrp1 (rabbit polyclonal)	Rio lab generated rabbit polyclonal	[20]
anti-active-Dcp1 (rabbit polyclonal)	Cell Signaling Technology (1:100)	Cat #9578 RRID: AB_2721060
anti-HA Tag (mouse monoclonal)	Cell Signalling Technology	Cat #2367 RRID:AB_10691311
anti-phospho-eIF2α (D9G8) (rabbit monoclonal)	Cell Signalling Technology	Cat #D9G8 #3398 RRID:AB_2096481
anti-β-Galactosidase (mouse monoclonal)	DSHB (1:100)	Clone ID/Product Name: JIE7 RRID: AB_528101
anti-β-Galactosidase (mouse monoclonal)	DSHB (1:100)	Clone ID/Product Name: mAb40-1a RRID:AB_528100
Goat anti-Mouse IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 488	Thermo Fisher Scientific	Cat: A-11001 RRID:AB_2534069
Goat anti-Rabbit IgG (H+L) cross-adsorbed secondary antibody, Alexa Fluor 488.	Thermo Fisher Scientific	Cat: A-11008 RRID:AB_143165
Goat anti-Mouse IgG (H+L) cross-adsorbed secondary antibody, Alexa Fluor 647.	Thermo Fisher Scientific	Cat:A-21235 RRID:AB_2535804
Goat anti-Rabbit IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 647	Thermo Fisher Scientific	Cat:A-21244 RRID:AB_2535812
Goat anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 555	Thermo Fisher Scientific	Cat: A-21424 RRID:AB_141780
Goat anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 555	Thermo Fisher Scientific	Cat: A-21429 RRID:AB_2535850
Anti-Guinea Pig-HRP	Jackson Immunoresearch	N/A
Anti-Guinea Pig-555	Mol. Probes/Invitrogen	N/A
anti-mouse-HRP	Jackson Immunoresearch	N/A
anti-beta tubulin E7 (mouse monoclonal)	DSHB (1:100)	RRID: AB_528499
IRDye 680RD Goat anti-Rabbit IgG	LICOR bio	Cat# 926-68071, RRID:AB_10956166
Biological samples		
Drosophila melanogaster.		
Chemicals, peptides, and recombinant proteins		

TRIzol® Reagent	Thermo Fisher Scientific	Cat#15596-026
Random Hexamers (50 µM)	Invitrogen	N8080127
Protoscript II Reverse Transcriptase	New England BioLabs	M0368L
RNase OUT Recombinant Ribonuclease Inhibitor	Invitrogen	10777-019
dNTP set 100 mM solutions	Thermo Fisher Scientific	R0181
oligo(dT)18 mRNA primer	New England BioLabs	S1316S
KAPA SYBR® FAST qPCR mix	KAPA Biosystems	Cat# KK4602
BSA (Bovine Serum Albumin)	Sigma	A8022
Triton X-100	Sigma	T8787
MgCl2	Sigma	M9272
EGTA	Sigma	E3889
Pipes	Sigma	P8203
Na2HPO4	Sigma	S5136
NaH2PO4	Sigma	S9638
Sodium Chloride	Sigma	S9888
Ethanol	Sigma	E7148
Isopropanol	Sigma	I9516
Paraformaldehyde, 16% w/v aq. soln., methanol free 10x10ml	Thermo Fisher Scientific	43368.9M
Critical commercial assays		
Immobilon® Crescendo Western HRP substrate	Millipore	Cat#WBLUR0500
Deposited data		
mRNA-Seq raw data	Lee et al [15]	GSE112864
mRNA-Seq raw data	Ji et al [21]	GSE124924
Proteomics dataset of *Drosophila* wing discs from wild-type, RpS3$^{+/-}$ mutant and Xrp1$^{+/-}$, RpS3$^{+/-}$ mutant larvae.	This study	PXD076715
Experimental models: Organisms/strains		
D. melanogaster. RpS12 mutation rpS12G97D	N. Baker [23]	Flybase: FBal0193403
D. melanogaster. UAS-RpS12 transgenic strain	N. Baker [22]	Flybase: FBal0337979
D. melanogaster. UAS-RpS12[G97D} transgenic strain	N. Baker [22]	N/A
D. melanogaster. act>CD2>Gal4, UAS-GFP recombinant expressing strain (this is the strong act>Flp-out cassette line on 3rd chromosome)	N. Baker [15]	Flybase: FBti0012408
D. melanogaster. act>y+>Gal4, UAS-CD8-GFP recombinant expressing strain (this is the weak act>Flp-out cassette line on the 2nd chromosome)	C. Delidakis, originally from Isabel Guerrero	N/A
D. melanogaster. act>Gal4, UAS-CD8-GFP recombinant expressing strain (on the 2nd chromosome)	Derived from tubGal80, FLP19A; act>Gal4, UAS-CD8-GFP/CyO of C. Delidakis	N/A
D. melanogaster. *nub-Gal4, UAS-RFP/CyO strain*	Kiparaki et al [19]	(Derived from FBst0063148)
D. melanogaster, RpS18 mutation M(2R)56f	N. Baker [15]	Flybase: FBal0011916
D. melanogaster mutant for RpL27A Df(2L)M24F11	N. Baker [15]	Flybase: FBab0001492

D. melanogaster endogenously HA-tagged Xrp1^{HA}allele	N. Baker [18]	Flybase: FBal0346068
D. melanogaster mutant Xrp1^{M2-73} allele	N. Baker [17]	Flybase: FBal0346068
D. melanogaster mutation RpS3^{NB}	N. Baker [15]	Flybase: FBal0346176
D. melanogaster transgenic strain Xrp1[02515]	N. Baker [15]	Flybase: FBal0009448
D. melanogaster transgenic strain en-Gal4	N. Baker [15]	RRID:BDSC_6356
D. melanogaster, RpS17 mutation M(3L)67C4	N. Baker [15]	Flybase: FBal0011935
D. melanogaster mutation *Xrp1^{08}*	K. Basler [16]	Flybase: FBal0387382
D. melanogaster. UAS-PPP1R15 transgenic strain	Bloomington Drosophila Resource Center	RRID:BDSC_76250
D. melanogaster. UAS-Xrp1^{long} transgenic strain	E. Storkebaum[47]	
D. melanogaster. UAS-Xrp1^{short} transgenic strain	E. Storkebaum[47]	
D. melanogaster. UAS-Xrp1^{long} AT-HOOK mutants and HA tagged transgenic strain	E. Storkebaum[47]	
D. melanogaster. UAS-Xrp1^{long} AT-HOOK mutants and HA tagged transgenic strain	E. Storkebaum[47]	
D. melanogaster Xrp1$^{Ex\text{-}long}$ deletion allele, here termed *Xrp1*$^{UAS\text{-}short}$, to emphasize UAS-driven expression of the Xrp1-RD short transcript.	E. Storkebaum[47]	FBal0344645 RRID:BDSC_607249
D. melanogaster Xrp1^{KO} deletion allele	E. Storkebaum[47]	RRID:BDSC_607250
D. melanogaster transgenic strain UAS-Irbp18 RNAi	Bloomington Drosophila Resource Center	RRID:BDSC_33652
D. melanogaster transgenic strain UAS-w RNAi	Bloomington Drosophila Resource Center	RRID:BDSC_33623
D. melanogaster transgenic strain UAS-Xrp1 RNAi	Vienna Drosophila Resource Center	VDRC:v107860 FLYBASE: FBst0479673
D. melanogaster transgenic strain UAS-Syp RNAi	Vienna Drosophila Resource Center	VDRC: v110542 FLYBASE: FBst0482109
D. melanogaster transgenic strain UAS-Syp RNAi	Vienna Drosophila Resource Center	VDRC: v33012 FLYBASE: FBst0459886
D. melanogaster transgenic strain UAS-TAF1B RNAi	Vienna Drosophila Resource Center	VDRC: v105873 FLYBASE: FBst0477699
D. melanogaster transgenic strain UAS-eIF6 RNAi	Vienna Drosophila Resource Center	VDRC: v108094 FLYBASE: FBst0479906
D. melanogaster UAS-RpL13A.Flag transgenic strain	Bloomington Drosophila Resource Center	RRID:BDSC_83684
D. melanogaster UAS-RpS17 transgenic strain	Fly ORF[92]	FlyORF: F001268 FLYBASE: FBst0500004
D. melanogaster UAS-GFP-RpL10Ab transgenic strain	Bloomington Drosophila Resource Center	RRID:BDSC_42683
Oligonucleotides		

RA_Xrp1_F:CCGGTGTTCATTGATGAATAC	Macrogen Europe	N/A
RC_Xrp1_F:CTTTCTAACGGTCAATTAAAAAATACC	Macrogen Europe	N/A
Xrp1_R: GTCAAGAATATCCGTCTCAG	Macrogen Europe	N/A
aTub84B_F: CACACCACCCTGGAGCATTC	Macrogen Europe	N/A
aTub84B_R: CCAATCAGACGGTTCAGGTTG	Macrogen Europe	N/A
Xrp1_Exon4_F: GGGAACTTGTGTCGACGACTC	Macrogen Europe	N/A
Xrp1_Exon4_R: CCGGACATCGATGACATGTCG	Macrogen Europe	N/A
Software and algorithms		
GraphPad Prism (v8)	GraphPad	RRID: SCR_002798 https://www.graphpad.com
Vast-tools v2.5.1	Tapial, J. et al., (2017)[93]	https://github.com/vastgroup/vast-tools
R (v4.2.1)	RCore Team (2022)	RRID:SCR_001905
DIA-NN, v1.8.1	Demichev et al., (2020) [94]	RRID:SCR_022865
Perseus, v2.0.5.0	Tyanova et al.,2016[95]	RRID:SCR_015753
ImageJ, V1.53t	NIH	RRID:SCR_003070 (https://imagej.nih.gov/ij/)
FastqQC (v0.12.1)	www.bioinformatics.babraham.ac.uk/projects/fastqc	RRID:SCR_014583
MultiQC (v1.25.1)	Ewels, P et al, 2016 [96]	RRID:SCR_014982
STAR's 2-pass mapping (v2.7.10)	Dobin, A et al, [97]	RRID:SCR_004463
Samtools' view (v1.17)	Danecek, P et al, [98]	RRID:SCR_002105
ggplot2 v3.4.2	Wickham et al. (2016) [99]	RRID:SCR_014601
ggfortify v0.4.16	Tang et al. (2016) [100]	N/A
BioRender	https://www.biorender.com	RRID:SCR_018361
DAVID	Huang et al. 2007[101]	RRID:SCR_001881
Adobe Photoshop CS5	Adobe systems	N/A
ZEISS Zen lite	Zeiss	https://www.zeiss.com/microscopy/en/products/software/zeiss-zen-lite.html
Salmon software (v0.14.1)	Patro et al. [102]	https://github.com/COMBINE-lab/salmon

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Species: Drosophila melanogaster.

Drosophila strains were generally maintained at 25^{o}C on standard wheat flour-sugar food supplemented with soy flour and CaCl2, at 25°C with 50%–70% relative humidity on a 12-hour light/dark cycle. Sex of third instar larvae dissected for most imaginal disc studies was not differentiated. Previous work reported no difference in the frequency of cell competition-induced apoptosis between male and female larvae[103]. The elimination of Xrp1-overexpressing clones was evident in both sexes. We therefore did not stratify analyses by sex in the present study.

Full genotypes for all the experiments are listed in Supplementary Table S2. The following genetic strains were used: UAS-RpL13A.Flag (Bloomington 83684), UAS-RpS17 (FlyORF, F001268), UAS-GFP-RpL10Ab (Bloomington 42683), act>y+>Gal4, UAS-CD8-GFP/Cyo (gift from Christos Delidakis), UAS-SypRNAi (v110542 and v33012), UAS-Xrp1$^{\text{short}}$-mut-HA [47], UAS-Xrp1$^{\text{long}}$-mut-HA [47], UAS-Xrp1$^{\text{short}}$ [47], UAS-Xrp1$^{\text{long}}$ [47], Xrp1$^{\text{KO}}$ [47], Xrp1$^{\text{Ex-long}}$ [47]. Other stocks have been described previously[15,18,19,22]. Xrp1 protein expression in some experiments were examined making use of an *Xrp1* allele tagged with HA (Xrp1-HA) at the endogenous *Xrp1* locus[18].

For the generation of the *Xrp1*$^{\textit{Ex}\text{-long}}$ allele, the transposon insertion lines PBac[WH]f05721 and P[XP]Xrp1d04790 were used following the general recombination scheme described by Parks et al.[47,53]. In the resulting *Xrp1*$^{\textit{Ex}\text{-long}}$ allele, approximately two-thirds of the Xrp1 locus is deleted, abolishing the expression of all Xrp1$^{\text{long}}$ transcripts while retaining the Xrp1-RD transcript[47] (Figure S3A). According to the recombination result *Xrp1*$^{\textit{Ex}\text{-long}}$ has an enhancer element containing 14 Gal4 UAS-binding sites upstream of the Xrp1-RD locus[51-53]. UAS sequences, which are near-palindromic motifs (CGGAGTACTGTCCTCCG), can act as orientation-independent enhancers when positioned upstream of a promoter[104,105].

METHOD DETAILS

Immunohistochemistry and Antibody Labeling

For antibody labeling, imaginal discs from late 3rd instar larvae were dissected in 1xPBS buffer and fixed in 4% formaldehyde in 1 x PEM buffer (1x PEM:100 mM Pipes, 1 mM EGTA, 1 mM MgCl2, pH 6.9). Fixed imaginal discs were 3x washed in PT (0.2% Triton X-100, 1xPBS) and blocked for at least 1hr in PBT buffer (0.2% Triton X-100, 0.5% BSA, 1 x PBS). Discs were incubated in primary antibody in PBT overnight at 4°C, washed three times with PT for at least 10 min each. The samples were incubated in secondary antibody in PBT overnight at 4°C (or for 3–4 hr at room temperature), and washed three times with PT for 5–10 min. After washes, samples were rinsed in 1 x PBS and the samples were incubated with the NuclearMask reagent (Thermofisher, H10325) for 10–15 min at room temperature. After washing 2 x with 1x PBS the imaginal discs were mounted in VECTASHIELD antifade mounting medium (Vector Laboratories, H-1000).

We used the following antibodies for staining: mouse anti- b-Galactosidase (J1e7, DSHB), rabbit anti-active-Dcp1 (cDcp1, Cell Signaling Techonology Cat#9578, 1:100), anti-phospho- eIF2α (D9G8) (rabbit monoclonal, Cell Signaling Technology), anti-HA Tag mouse monoclonal (Cell Signalling Cat#2367), anti-beta galactosidase (mAb40-1a) (mouse monoclonal), guinea pig anti-Xrp1[29]. Secondary Antibodies were Alexa Fluor conjugates (Thermofisher).

Clonal analysis

Genetic mosaics were generated using the FLP/FRT system employing heat shock-inducible FLP (hsFLP) transgenic strains[106,107]. For making clones through mitotic recombination using heat shock-inducible flipase recombinase (hsFLP), $Rp^{+/-}$ larvae were subjected to 30 min heat shock at 37 °C, 60 ± 12 hours after egg laying (AEL) and dissected 68-72 hr later. For making clones by excision of a FRT cassette, larvae were subjected to 20–60 min heat shock at 37 °C, 40 ± 12 AEL and dissected 68-72 hr later. Notice that longer heat shock increase FRT/FLP recombination because the hs-FLP promoter drives more FLP expression for a longer time, which raises the chance of FRT sites recombine in more cells. Thus, extending the heat-shock period from 25 min to 45 min increases loser clone frequency and mosaic size. Full genotypes for all figures and heat shock times are listed in Supplementary Table S2.

Image Acquisition and Processing

Confocal images were recorded using Leica Sp8 and Zeiss confocal microscopes using 20x and 40x objectives. Images were processed using Leica's and Zeiss software, and Adobe Photoshop CS5 Extended.

Alternative splicing analysis

Splicing analysis of the RNA-Seq data (GEO accession numbers: GSE112864 and GSE124924) was performed using the Alternative Splicing and Transcript Tools (VAST-TOOLS) v2.2.2 (https://github.com/vastgroup/vast-tools), using Drosophila genome release dm6[93]. For all events, a minimum read coverage of 10 actual reads per sample was required, as described[108]. PSI values for single replicates were quantified for all types of alternative events, including single and complex exon skipping events (S, C1, C2, C3, ANN), alternative donor and acceptor splice choices (Alt5 and Alt3, respectively) and retained introns (IR-S, IR-C). A minimum dPSI of 5% was required to define differentially spliced events upon each mutant, along with a *p-value* < 0.05 calculated using 2-tailed unpaired Wilcoxon Mann–Whitney test. Sashimi graph was generated using IGV and Aligned BAM files (dm6 genome).

Salmon isoform quantification analysis

The same RNA-seq data was quantified using the Salmon software (v0.14.1)[102]. Transcript abundance was estimated using quasi-mapping-based mode. The analysis was performed using the pre-built transcriptome index generated from the Drosophila melanogaster genome (BDGP6.46 from Ensembl release 112), while setting the minimum score fraction parameter at default setting 0.65. The output TPM values were further processed and normalized using downstream analysis tool tximport[109](v1.24.0) in R (v4.2.1)[110].

RA and RG isoforms differ only by a ~150-nt segment in the 3′ UTR. Short-read quantifiers (Salmon) cannot reliably distinguish isoforms that differ solely at transcript ends unless reads overlap the unique region. We therefore observe that absolute counts are sensitive to parameter stringency, e.g. after applying --minScoreFraction 0.95, while the fold-change direction across conditions remains stable. Our short-read data should thus be interpreted as robust for relative changes at the Xrp1 locus but conservative for the RA:RG split in absolute terms. Definitive allocation between RA and RG would require long-read–guided references or targeted 3′ assays, which were beyond the scope of this work.

Targeted RNA-seq read counting for allele-specific detection of the *Xrp1$^{M2\text{-}73}$* mutation

FASTQ raw reads were quality checked using FastqQC (v0.12.1) [ref:https://www.bioinformatics.babraham.ac.uk/projects/fastqc] and MultiQC(v1.25.1)[96]. RNA-seq reads were aligned to the reference genome of Drosophila melanogaster [UCSC version dm6[111] using STAR's 2-pass mapping (v2.7.10)[97]. The optional flags used in the 2-pass mapping are shown in Supplementary Table S3. The number of reads matching a sequence of interest flagging the mutation of interest was counted based on properly paired uniquely aligned reads using Samtools' view (v1.17)[98]. The mutation of interest was *Xrp1$^{M2\text{-}73}$*, located in chr3R:18925491[15]. The sequence to match was ATTCATCACGTAGGAGCCCCT for the mutation and ATTCATCACGGAGGAGCCCCT for the wild type, which included the mutation site and 10 flagging nucleotides in both directions.

Sample collection, RNA isolation, cDNA synthesis and RT-PCR/qPCR

RNA isolation from drosophila wing discs was performed with the TRIzol® reagent (Invitrogen, Cat. No 15596026), according to manufacturer's instructions as described previously [15].

Reverse transcription (RT) was carried out with 1μg of total RNA, using random hexamer primers (Invitrogen), OligodT (New England BioLabs, Inc.), RNaseOUT™ Recombinant Ribonuclease Inhibitor (Thermo Fisher Scientific) and Protoscript II reverse transcriptase (New England BioLabs, Inc.), according to manufacturer's instructions.

Semi-quantitative PCR reactions (RT-PCR) were carried out with 2 μl of cDNA, with the following primers for the Xrp1 DmeEX0007032 splicing event: Xrp1_Exon4_F: GGGAACTTGTGTCGACGACTC and Xrp1_Exon4_R: CCGGACATCGATGACATGTCG. PCR conditions included 30 cycles of denaturation at 94^{o}C for 30'', annealing at 60^{o}C for 30'' and extension at 72^{o}C for 90''. Inclusion/exclusion bands for these events were analyzed by 10% acrylamide electrophoresis and quantified using ImageJ (v1.53t).

Quantitative PCR (qPCR) was performed in a Corbett Rotor Gene™ RG 6000 using Kapa SYBR®FAST (KAPA Biosystems) qPCR mix. The following primer sequences were used: RA_Xrp1_F:CCGGTGTTCATTGATGAATAC, RC_Xrp1_F:CTTTCTAACGGTCAATTA AAAAATACC, Xrp1_R: GTCAAGAATATCCGTCTCAG. As reference gene, aTub84B was used with the following primers: aTub84B_F: CACACCACCCTGGAGCATTC and aTub84B_R: CCAATCAGACGGTTCAGGTTG. qPCR conditions included 30 cycles of denaturation at 94oC for 15’’, annealing at 58oC for 15’’ and extension at 72oC for 15’’.

Western blot analysis

Protein lysates from7 (using the rabbit anti-Xrp1 antibody) or 30-40 (using the guinea pig anti-Xrp1 antibody) wing imaginal discs of third instar larvae of each genotype were separated on SDS–polyacrylamide gels (11%). After blocking, we incubated with the primary anti-Xrp1 antibodies [rabbit(1:1000) or guinea pig(1:2000)]. , Ubiquitous depletion of Syp with actin-Gal4 driver caused embryonic lethality. Therefore, we retrieved lysates from wing discs where the weak Syp-RNAi line (v33012) was clonally expressed. For the rabbit anti-Xrp1 detection we used the secondary antibody anti-rabbit conjugated with IRDye 680 nm (LI-COR) and imaged on a LI-COR Odyssey scanner system. For the guinea pig anti-Xrp1 antibody we used HRP conjugated antibody and proteins were visualized with chemiluminescence (Immobilon Crescendo, Millipore). Extracts from flies overexpressing N-terminal HA-tagged variants of either UAS-Xrp1^{long} or UAS-Xrp1^{short}][47], both carrying mutations in their AT-hook motif, revealed that Xrp1 isoforms migrate slower in SDS-PAGE than their predicted size, consistent with previous observations [Igaki T, personal communication and Mallik et al.[47]].

Proteomic sample preparation using the Sp3-mediated protein digestion protocol

For proteomic analysis we used three to four biological replicas from each genotype. From each genotype wing imaginal discs of third-instar wandering larvae were dissected in 0.1 M sodium phosphate buffer (pH 7.2). We dissected as many discs we could within 10–15 min at room temperature. Drosophila tissue was excised and homogenized in a lysis buffer consisting of 4% SDS, 0.1M DTT, 0.1M TEAB and lysed by incubation at 99 ^{o}C for 5 min. Samples were immediately frozen at -80 ^{o}C. In total, 50 imaginal discs were used for each biological sample. After thawing, sample were sonicated in waterbath, followed by a centrifugation step for 15 min at 17000xg.

The lysed samples were processed according to the Sp3 protocol (Hughes) including an alkylation step in 200 mM iodoacetamide (Acros Organics). 20 ug of beads (1:1 mixture of hydrophilic and hydrophobic SeraMag carboxylate-modified beads, GE Life Sciences) were added to each sample in 50% ethanol. Protein clean-up was performed on a magnetic rack. The beads were washed two times with 80% ethanol and once with 100% acetonitrile (Fisher Chemical). The captured on beads proteins were digested overnight at 37o C under vigorous shaking (1200 rpm, Eppendorf Thermomixer) with 1 ug Trypsin/LysC (MS grade, Promega) prepared in 25 mM ammonium bicarbonate. Next day, the supernatants were collected and the peptides were purified using a modified Sp3 clean up protocol and finally solubilized in the mobile phase A (0.1% formic acid in water), sonicated and the peptide concentration was determined through absorbance at 280 nm measurement using a nanodrop instrument.

LC-MS/MS Analysis

Samples were run on a liquid chromatography tandem mass spectrometry (LC-MS/MS) setup consisting of a Dionex Ultimate 3000RSLC online with a Thermo Q Exactive HF-X Orbitrap mass spectrometer. Peptidic samples were directly injected and separated on an 25 cm-long analytical C18 column (PepSep, 1.9μm3 beads, 75 μm ID) using an 60 minutes long run. A full MS was acquired in profile mode using a Q Exactive HF-X Hybrid Quadrupole-Orbitrap mass spectrometer, operating in the scan range of 375-1400 m/z using 120K resolving power with an AGC of 3x 106 and max IT of 60ms followed by data independent analysis using 8 Th windows (39 loop counts) with 15K resolving power with an AGC of 3x 105 and max IT of 22ms and a normalized collision energy (NCE) of 26. Each biological sample was analyzed in three technical replicas.

Proteomic database search

Orbitrap raw data was analyzed in DIA-NN 1.8.1 (Data-Independent Acquisition by Neural Networks)[94] through searching against the reference proteome of *Drosophila melanogaster* (13767 entries) retrieved from Uniprot database in the library free mode of the software, allowing up to two tryptic missed cleavages. A spectral library was created from the DIA runs and used to reanalyse them. DIA-NN default settings have been used with oxidation of methionine residues and acetylation of the protein N-termini set as variable modifications and carbamidomethylation of cysteine residues as fixed modification. N-terminal methionine excision was also enabled. The match between runs (MBR) feature was used for all analyses and the output (precursor) was filtered at 0.01 FDR and finally the protein inference was performed on the level of genes using only proteotypic peptides.

QUANTIFICATION AND STATISTICAL ANALYSIS

Each experiment was performed independently at least three times.

Each data point in the plots represents an independent biological replicate

Details of statistical evaluations and the sample sizes were written in the Figure legends.

Statistical analyses and plots of qPCR and Alternative Splicing analysis

Statistical tests were performed as indicated in figures legends using R (v4.2.1) or GraphPad Prism (version 8). Principal Component Analysis (PCA) was performed on PSI values of all the exonic splicing events and visualized using ggfortify[100](v0.4.16) and ggplot2 (v3.4.2) packages in R[99] (v4.2.1).

Proteomic statistical analysis

For statistical and bioinformatics analysis, as well as for visualization, Perseus, which is part of Maxquant suite, was used[95]. The protein intensities were transformed to logarithmic values [log2(x)]. The protein groups were filtered to obtain at least 2 valid values in at least one group and missing values were imputed from normal distribution. The label-free quantified proteins were subjected to statistical analysis using both ANOVA test and student t-tests, both with permutation-based FDR calculation. The statistical test results were visualized in volcano plot displaying the difference between the two samples expressed as log2(x) versus their statistical significance expressed as −Log10(p-value). Hierarchical clustering was carried out on Z-score transformed LFQ values using average linkage of Euclidian distance. GO Enrichment analysis for biological processes was performed using DAVID functional annotation tools[101] with Flybase Gene ID as identifiers, the Drosophila background and the GOTERM_DIRECT annotation categories[112,113].

SUPPLEMENTAL INFORMATION

Document S1: Figures S1-S9 and supplemental references

Supplementary Table S1: PSI values for Xrp1

Supplementary Table S2: Full genotypes for all the experiments, heat shock times and N number for each experiment.

Supplementary Table S3: Optional flags used in STAR's 2-pass mapping

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Supplemental Figures and Legends

Figure S1. Related to Figure 1.

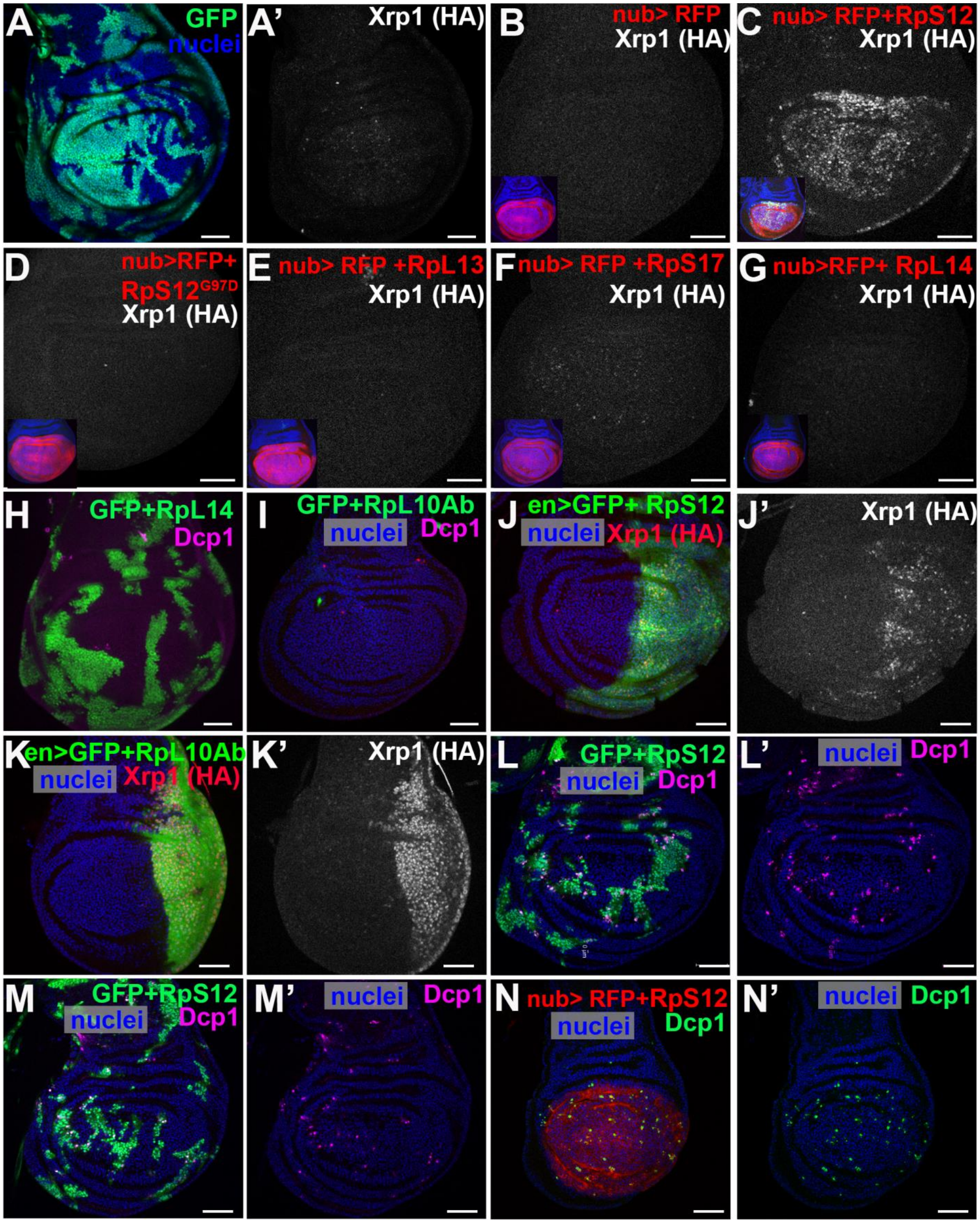


Figure S1 RpS12, but not other RPs, triggers Xrp1-dependent cell competition in wild-type cells.

Confocal images of third-instar wing discs of the indicated genotypes are shown. Panels H-I show projections of the central disc-proper region. All other panels show single confocal planes. anti-HA staining shows Xrp1-HA expression from the endogenously HA-tagged *Xrp1* allele, while active cDcp1 staining marks apoptotic cells. Nuclei are in blue. Scale bars, 50μm.

(A) Clones overexpressing GFP in wild-type discs do not induce Xrp1-HA expression (n=6).

(B-G) Xrp1- HA expression in wing discs where nub-Gal4 drives co-expression of RFP together with the indicated UAS-driven transgenes. Insets show nubGal4, UAS-RFP expression in red and Xrp1-HA staining in green. In (C), overexpression of RpS12 strongly induces Xrp1- HA(n=12), whereas RFP alone (n=7) (B), or in combination with $RpS12^{G97D}$ (n=10) (D), RpL13 (n=5) (E), or RpL14 (n=10) (G) fails to induce Xrp1- HA. RpS17 overexpression induced only negligible Xrp1- HA levels (n=5) (F).

(H) Clonal overexpression of RpL14 together with GFP (green) in wild-type discs does not induce cell death (n=5) (cDcp1, magenta).

(I) Clones overexpressing RpL10Ab-GFP together with GFP do not survive (n=7).

(J) en-Gal4 drives co-expression of GFP and RpS12 in the posterior domain of a wild-type wing disc and induces Xrp1-HA expression (n=10).

(K) en-Gal4 drives co-expression of GFP and GFP-RpL10Ab in the posterior compartment and strongly induces Xrp1-HA and reduces compartment size (n=5).

(L,M) Additional examples of wild-type wing discs in which clonal RpS12 overexpression, together with GFP, induces apoptotic cell death (cDcp1, magenta) at the boundaries with the wild-type cells (n=10).

(N) Co-expression of UAS-RpS12 together with UAS-RFP (red) using the nub-Gal4 driver induces sporadic cell death (n=6) (cDcp1). Control similar to S5B.

Figure S2. Related to Figure 2

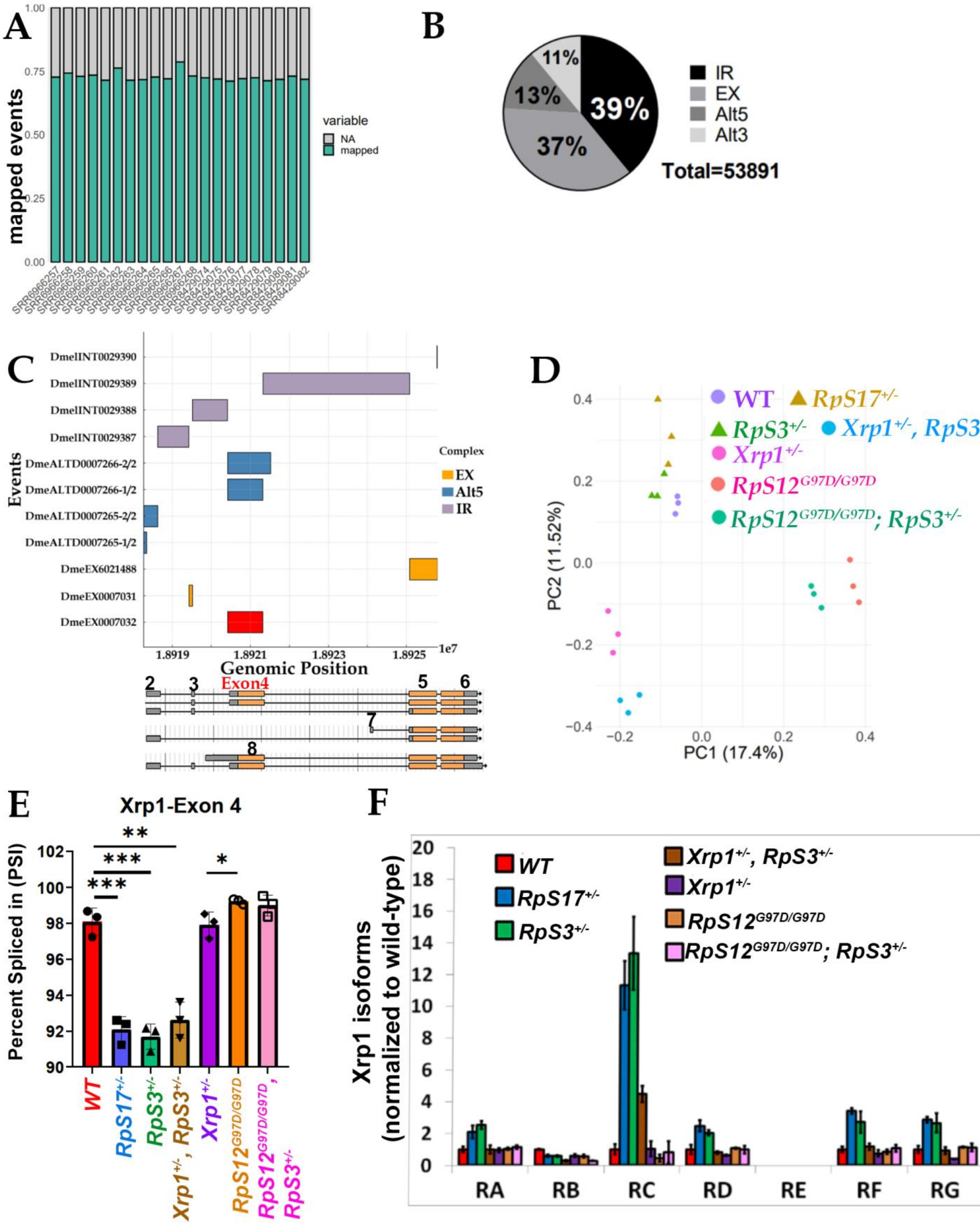


Figure S2. Alternative splicing analysis and Xrp1 isoform changes across genotypes.

(A) Bar graph showing the percentage of mapped alternative splicing events in all the individual RNA-seq samples analyzed. 3 biological samples were analyzed for each genotype.

(B) Pie-chart displaying the distribution of mapped alternative splicing events by event type, with distinct shading indicating intron retention (IR), alternative exon (EX), alternative 5' site (Alt5) and alternative 3' splice site (Alt3) categories. Across all genotypes, 53891 alternative splicing (AS) events were identified, with roughapproximately 40% corresponding to alternative exons and a similar fraction to retained introns.

(C) Bar graph illustrating the genomic locations of annotated alternative splicing events in *Xrp1* (y-axis). The x-axis shows the position on the chr3R chromosome and the y-axis shows the VastdB ID number for each AS event. Events are categorized into three groups based on type: EX (orange), Alt5 (blue), and intronic or IR (purple). Each event spans specific genomic region, represented by horizontal bars. The DmeEX0007032 event is highlighted in red. *Xrp1* gene locus is presented below to facilitate identification of the events.

(D) Principal Component Analysis (PCA) of PSI values for all mapped splicing events across the indicated genotypes. Each point corresponds to an individual sample. 3 biological samples were analyzed for each genotype.

(E) PSI values for *Xrp1* exon 4 inclusion in the final mRNA, shown as mean ± SEM. *p-values* are calculated by Student's t-test (** $p< 0.01$, *** $p< 0.001$). Genotypes are color-coded: wild-type (red), *RpS17*$^{+/-}$ (blue), *RpS3*$^{+/-}$ (green), *Xrp1*$^{+/-}$, *RpS3*$^{+/-}$ (brown), *Xrp1*$^{+/-}$ (purple), *RpS12*$^{G97D/G97D}$ (orange), and *RpS12*$^{G97D/G97D}$,*RpS3*$^{+/-}$ (pink). 3 biological samples were analyzed for each genotype.

(F) Fold change of the Xrp1 mRNA isoforms relative to wild-type levels. The graph shows the ratio of transcripts per million (TPM) for each isoform in the indicated genotypes, normalized to TPM of the same isoform in wild-type cells. Isoform E is not shown because its low abundance results in high SD. Genotypes and color coding are as in E. Error bars represent ±1 SD derived from RNAseq data. 3 biological samples were analyzed for each genotype.

Figure S3. Related to Figure 3

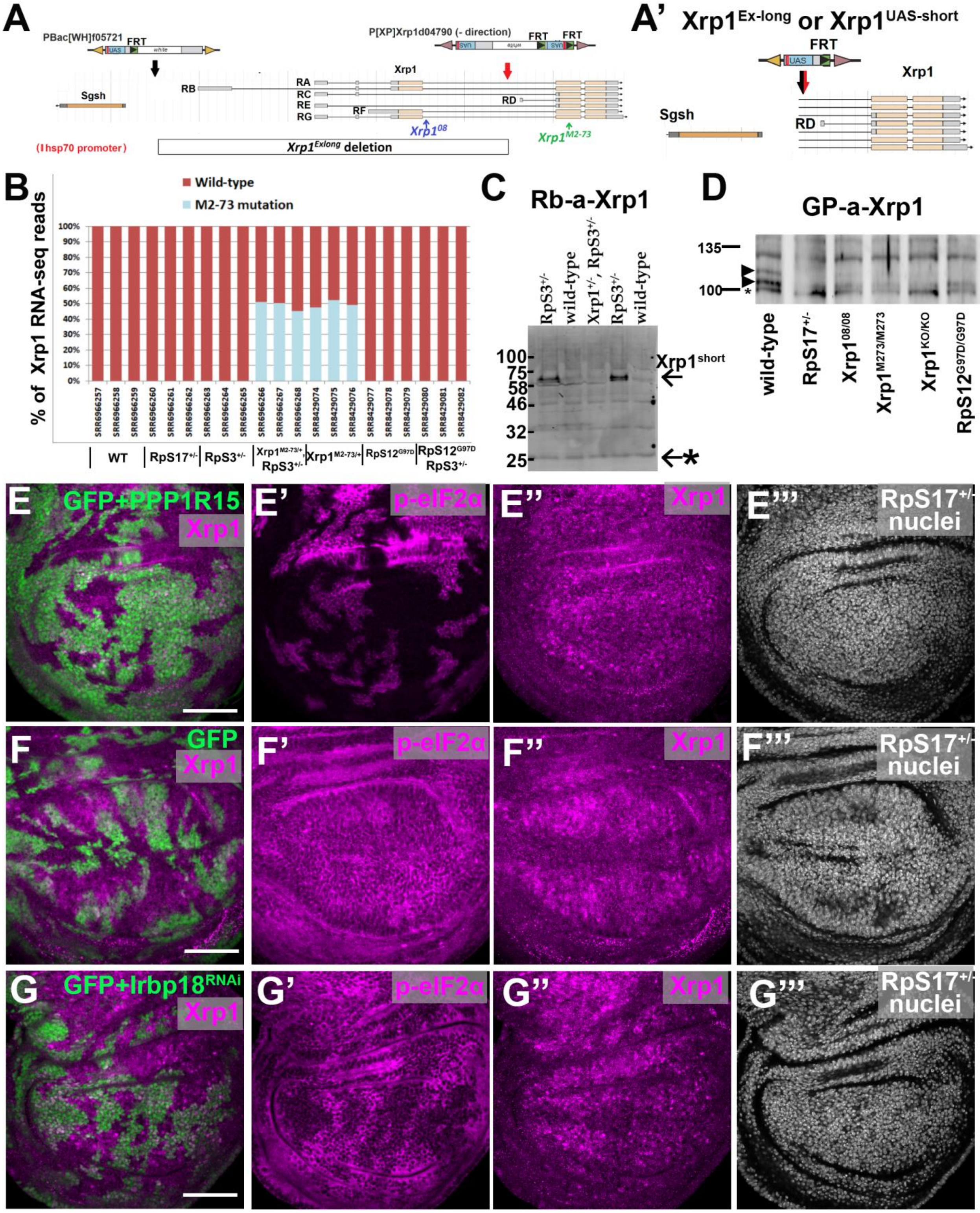


Figure S3. RpS12 induces Xrp1^{short} in *Rp*$^{+/−}$ cells, independently of p-eIF2α.

(A, A') Schematic view of the chromosome 3R region around the *Xrp1* locus (A), indicating the positions of the *Xrp1* mutation. *Xrp1*$^{M2\text{-}73}$ is a dominant-null allele that contains a stop codon within exon 5. *Xrp1*$^{M2\text{-}73/+}$ heterozygosity (depicted as *Xrp1*$^{+/−}$) prevents Xrp1 activation in *Rp*$^{+/−}$ cells and rescues *Minute* phenotypes, including cell competition [1-3]. The *Xrp1*^{08} mutation has an intronic mutation near exon 4 that disrupts a conserved splice enhancer[4]. The genomic segment deleted in the *Xrp1*$^{Ex\text{-}long}$ (referred to in our study as *Xrp1*$^{UAS\text{-}short}$) allele [5] is shown as a boxed region. In A', a schematic view of the *Xrp1*$^{UAS\text{-}short}$ allele is shown, containing upstream activating sequence (UAS) motifs immediately 5' to the deletion breakpoint. The promoter of Xrp1-RD transcript remains after the genomic deletion and drives expression of Xrp1$^{short\text{-}}$RD transcript. Also shown in the scheme is the hsp70 promoter of the PBac[WH]f05721 insertion line, which also remains after deletion.

(B) Percentage of allele-specific quantification of RNA-seq reads mapping to the wild-type (red) or *Xrp1*$^{M2\text{-}73}$ (blue) allele in wing discs of the indicated genotypes. RNA-seq genotypes include: (a) wild-type, (b) *RpS17*$^{+/−}$, (c) *RpS3*$^{+/−}$, (d) *Xrp1*$^{+/−}$*, RpS3*$^{+/−}$, (e) *Xrp1*$^{+/−}$, (f) *RpS12*$^{G97D/G97D}$, and (g) *RpS12*$^{G97D/G97D}$*,RpS3*$^{+/−}$. In both *Xrp1*$^{+/−}$*, RpS3*$^{+/−}$ and *Xrp1*$^{+/−}$ samples, the wild-type and M2-73 alleles are equally represented, indicating that the M2-73 nonsense mutation does not further engage nonsense-mediated decay (NMD) or reduce Xrp1 mRNA abundance or stability, despite Xrp1 being a nonsense-mediated decay (NMD) target in Drosophila[6]. Each genotype includes three biological replicates.

(C) Western blots of protein extracts from wing discs of *wild-type*, *RpS3*$^{+/−}$ and *Xrp1*$^{+/−}$*; RpS3*$^{+/−}$ larvae probed with a rabbit anti-Xrp1 antibody (Rio laboratory[7]). Only the short isoform is detected, whereas the long remains undetectable. The Xrp1^{short} isoform migrates at a higher molecular weight than predicted, as previously shown in Mallik et al.[5] and in personal communication with Dr Tatsushi Igaki. Xrp1^{short} levels are increased in *RpS3*$^{+/−}$ cells, and this induction is abolished in *Rp*$^{+/−}$ cells heterozygous for the dominant null *Xrp1*$^{M2\text{-}73}$ allele (*Xrp1*$^{+/−}$). Background bands (asterisk) serve as a loading control.

(D) Part of the western blot from Figure 3A probed with the guinea pig anti-Xrp1 antibody (Ryoo laboratory), focusing on the region corresponding to the Xrp1^{long} isoform. Xrp1^{long} is detected as a doublet, with the upper band likely representing a post-translationally modified form, whereas the lower band co-migrates with a background band that is also faintly visible in homozygous Xrp1 knockout (Xrp1$^{KO/KO}$) discs. Due to the low abundance of the upper band, sometimes only the lower band is detectable. Lanes: (1) *wild-type*, (2) *RpS17*$^{+/-}$, (3) *Xrp1*$^{08/08}$, (4) *Xrp1*$^{M2\text{-}73/M2\text{-}73}$, (5) *Xrp1*$^{KO/KO}$, (6) *RpS12*$^{G97D/G97D}$

(E-G) Confocal images of *RpS17*$^{+/-}$ third-instar wing discs of the indicated genotypes are shown. Panels show single confocal planes. Discs were co-stained with Rb-p-eIF2α (E'-G') and guinea-pig anti-Xrp1 (E''-G''). *RpS17*$^{+/-}$ nuclei are shown in E'''-G'''. Clones were induced by a 25-min heatshock. Scale bars, 50μm.

E) Clones overexpressing PPP1R15 and GFP in *RpS17*$^{+/-}$ wing discs show complete loss of p-eIF2α (E'), whereas Xrp1 levels remain unchanged (E'').

F) Control clones overexpressing GFP alone in *RpS17*$^{+/-}$ wing disc do not alter p-eIF2α (F') or Xrp1 (F'').

G) Clones overexpressing Irbp18-RNAi and GFP in *RpS17*$^{+/-}$ wing disc display reduced p-eIF2α (G') and Xrp1 (G'').

Figure S4. Related to Figure 4

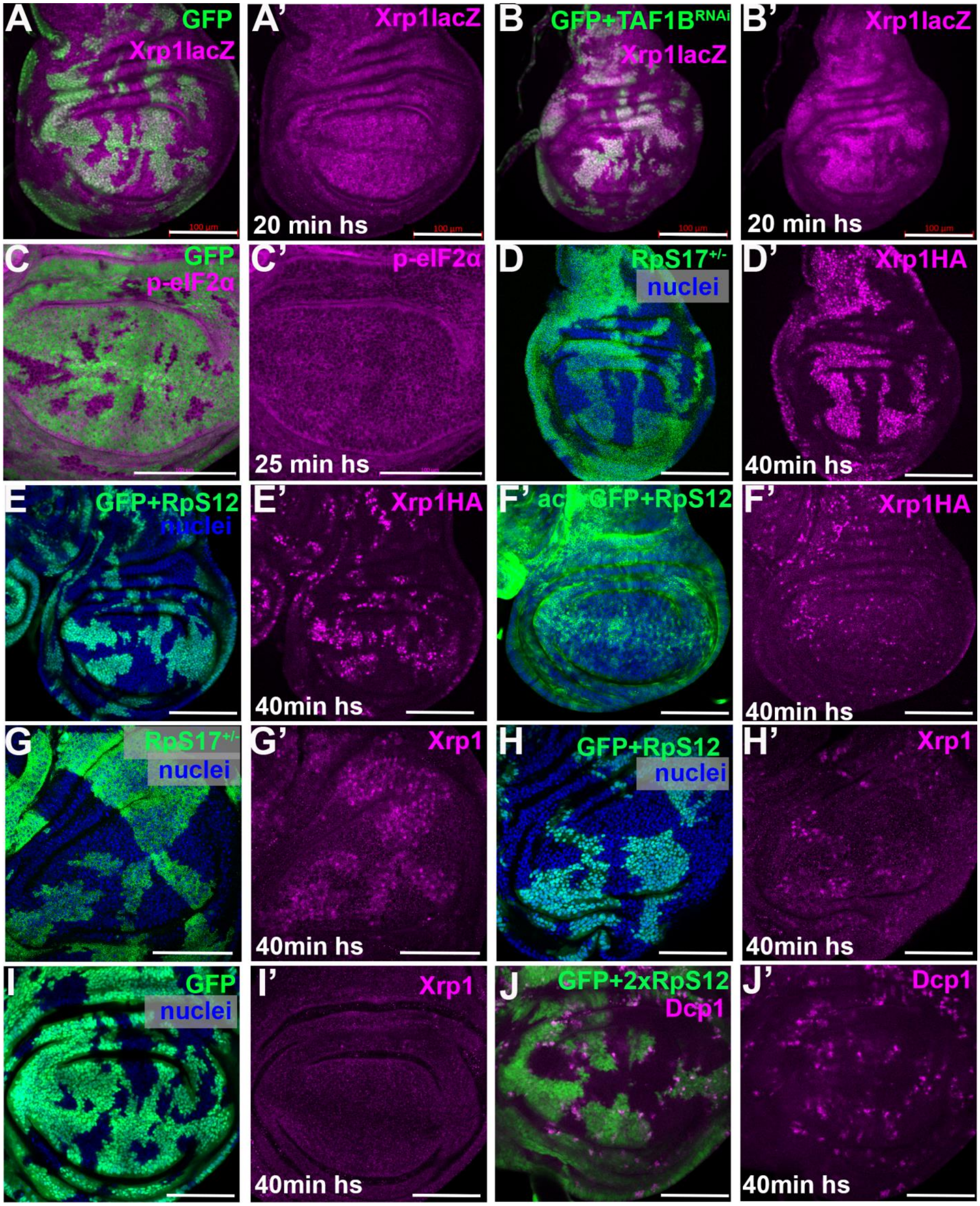


Figure S4. RpS12 induces Xrp1 and boundary cell death in a dosage-dependent manner in wild-type cells.

Confocal images of third-instar wing discs of the indicated genotypes are shown. In panel D, wild-type clones were induced in an *RpS17*$^{+/-}$ background by mitotic recombination and are labeled with anti-βgal (green). In all other panels, discs contain clones generated with the FLP-out Gal4 technique, marked with GFP, and overexpressing also the indicated transgenes. Heat-shock induction times for each experiment are shown. All panels are single confocal planes. Nuclei are in blue. Scale bars, 100μm.

(A) Clones overexpressing UAS-GFP in wild-type wing discs do not increase *Xrp1*-lacZ, labeled with anti-βgal shown in magenta. This serves as a negative control for Figures 4C-4D.

(B) Clones overexpressing TAF1B-RNAi in wild-type wing discs increase *Xrp1*-lacZ expression, labeled with anti-βgal shown in magenta. This serves as a positive control for Figures 4C-4D.

(C) Clones overexpressing UAS-GFP in wild-type wing discs do not increase p-eIF2α levels. This serves as a negative control for Figures 4E-4F.

(D-F) Immunostaining was performed in the same reaction tubes. Xrp1 protein expression in these experiments was examined making using the endogenous Xrp1 HA-tagged allele, labeled with anti-HA antibody. Clones of *RpS17*$^{+/-}$ cells (green in D) adjacent to wild-type cells present increased Xrp1-HA expression (D'). RpS12-overexpressing clones (green in E) also show increased Xrp1-HA expression (E'), but the pattern is less uniform than in *RpS17*$^{+/-}$ cells (D'). Ubiquitous expression of RpS12 using the act>Gal4 driver shows mild upregulation of the Xrp1-HA protein (F'). For this experiment negative control behaved as in Figures 1B and S1A-S1B, where Xrp1-HA expression is negligible in wild-type cells.

(G-I) Immunostaining of samples with the guinea-pig anti-Xrp1 antibody shown in (G) and (H) was performed in the same reaction; while sample (I) was processed in parallel as a negative control. Clones of *RpS17*$^{+/-}$ cells (green in G) adjacent to wild-type cells display increased Xrp1 levels (G'), while RpS12-overexpressing clones (green in H) show weaker and less uniform Xrp1 expression (H'). GFP-overexpressing clones (green in I) do not present Xrp1 upregulation.

(J) Two-copy (2x) RpS12 overexpression causes prominent boundary cell death (cDcp1, magenta). The increased heat-shock duration increased the clone size of RpS12-overexpressing cells, allowing clearer observation of boundary cell death.

Figure S5. Related to Figure 5

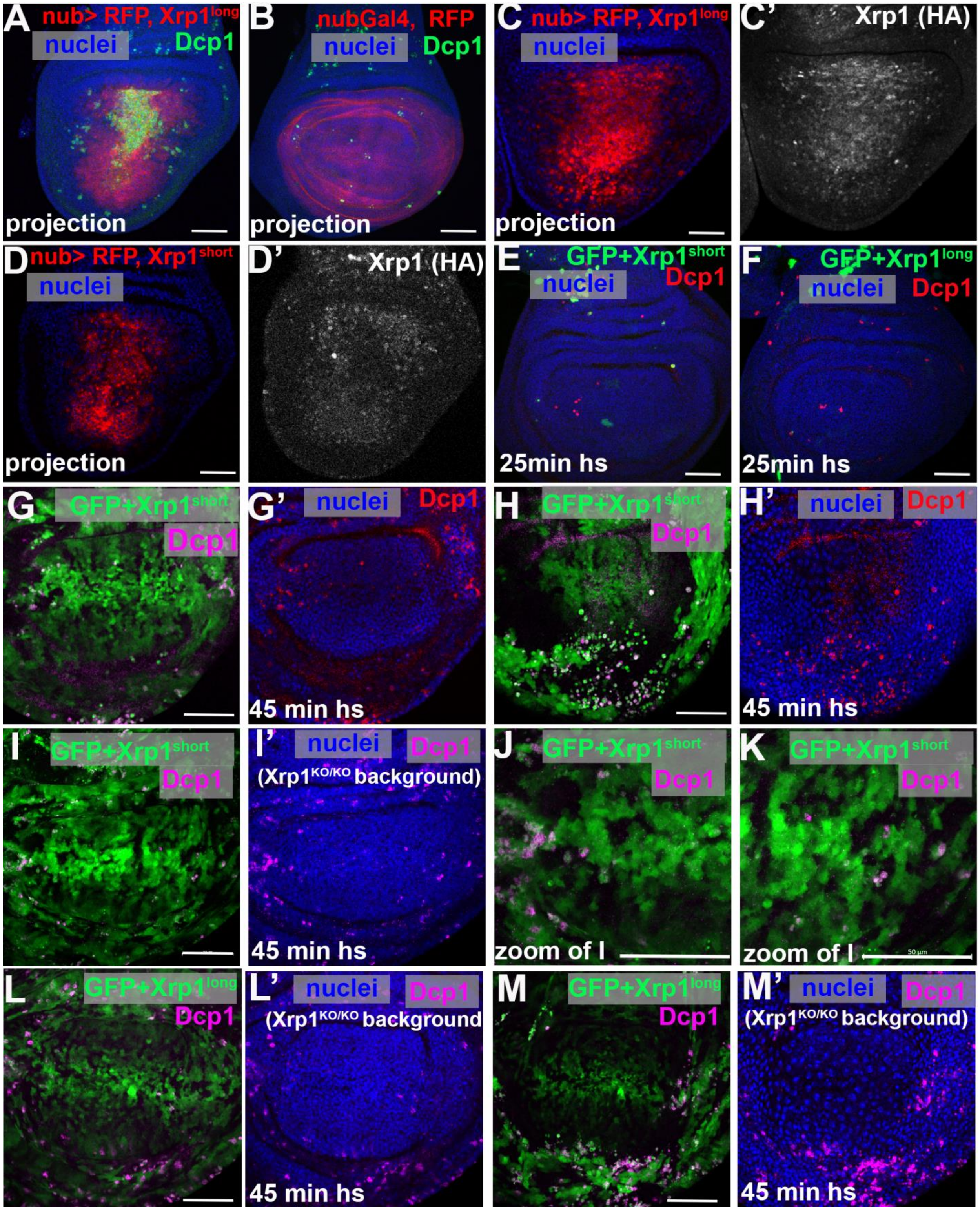


Figure S5 Xrp1 isoforms induce Xrp1 autoregulation and cell death in wing imaginal discs.

Confocal images of third-instar wing discs of the indicated genotypes are shown. Panel A-F show projections of Dcp1 staining in the central region of the disc-proper. Panel J-M show single confocal planes. In panels E-M clones were generated using the FLP-out Gal4 technique, where GFP was overexpressed together with the indicated Xrp1 transgenes. In panels E-M, active cDcp1 staining marks apoptotic cells, which is shown either in magenta or in red. Nuclei are shown in blue. Scale bars, 50µm.

(A) nub-Gal4 drives co-expression of the UAS-RFP reporter (red) together with UAS-Xrp1^{long} in the wing pouch. Xrp1^{long} overexpression results in massive cell death (Dcp1, green)

(B) nub-Gal4 drives the expression of the UAS-RFP reporter (red) in the wing pouch. Negligible Dcp1 staining is detected (Dcp1, green).

(C, D) nub-Gal4 drives the co-expression of RFP (red) together with Xrp1^{long} (C) and Xrp1^{short} (D) in the wing pouch, resulting in induction of the endogenously HA-tagged Xrp1 allele (C', D').

(E, F) Clones overexpressing either Xrp1^{short} (E) or Xrp1^{long} (F) (marked in green) do not survive (Dcp1, red). In these experiments the act>CD2>Gal4 cassette was used. Clones were induced by a 25 min heat shock.

In experiments shown in panels G-M the act>y+>Gal4 cassette was used to generate clones; this cassette is weaker than the act>CD2>Gal4 cassette used in panels E and F. Clones were induced by a 45-min heat shock.

(G, H) Clonal overexpression of the Xrp1^{short} isoform (marked in green) in wild-type disc exhibits sporadic boundary cell death in the disc proper (cDcp1). In the peripodial membrane of the same disc, Xrp1^{short}-overexpressing clones (marked in green) show prominent boundary cell death (cDcp1)(H).

(I-K) Clonal overexpression of the Xrp1^{short} isoform (marked in green) in a homozygous Xrp1^{KO} disc exhibit sporadic boundary cell death in the disc proper (cDcp1). (J) and (K) show higher magnification views of the disc where boundary cell death is more evident.

(L, M) Clonal overexpression of the $Xrp1^{long}$ isoform (marked in green) in a homozygous $Xrp1^{KO}$ disc exhibit sporadic boundary cell death in the disc proper (cDcp1)(L). In the peripodial membrane of the same disc, $Xrp1^{long}$-overexpressing clones show prominent boundary cell death (cDcp1) (M).

Figure S6. Related to Figure 5

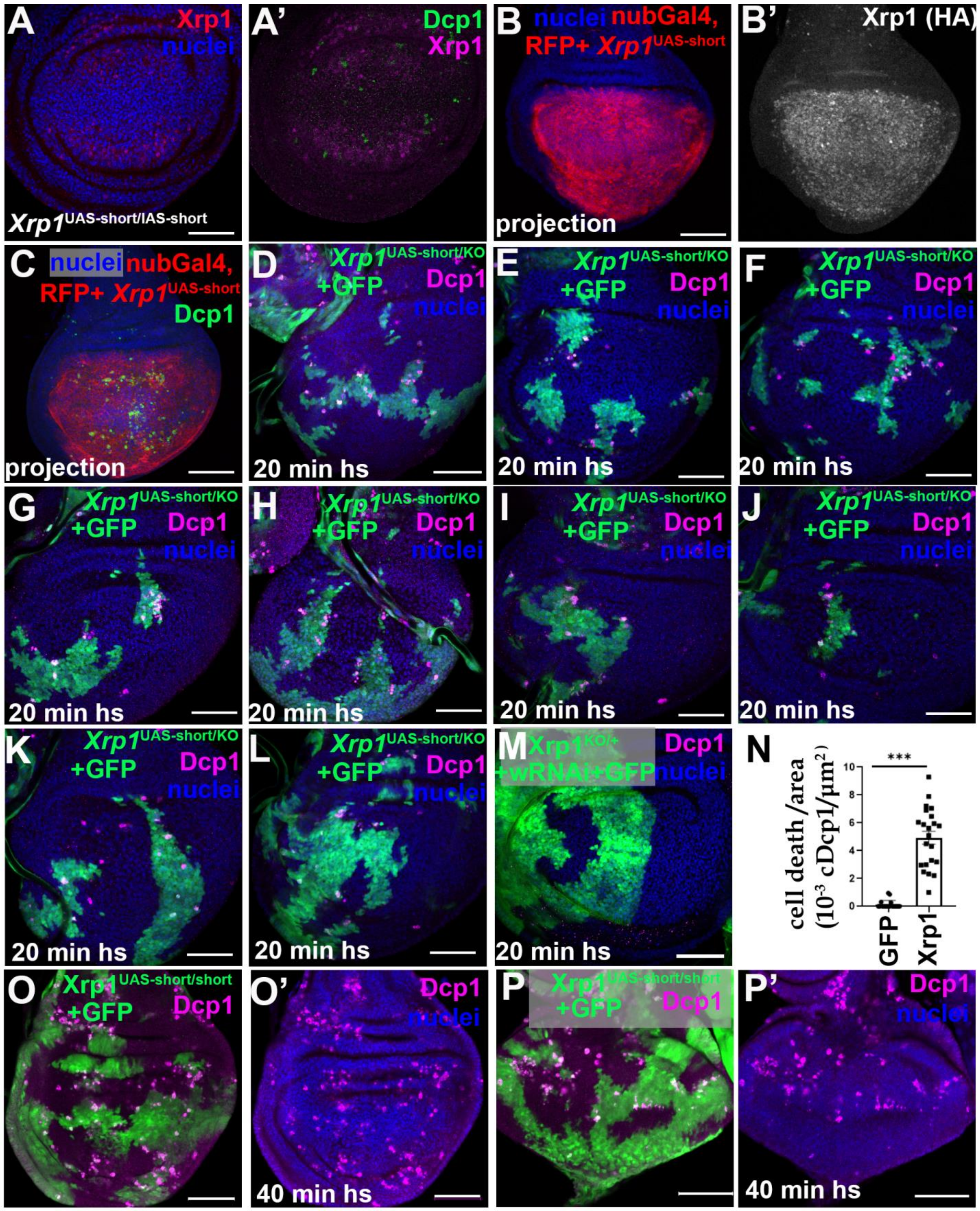


Figure S6. Clones overexpressing Xrp1^{short} from the *Xrp1*$^{UAS\text{-}Exlong}$ allele show boundary cell death.

Confocal images of third-instar wing discs. Panels B and C show projections from the disc-proper region. All other panels show single confocal planes. Scale bars, 50µm.

(A) Discs homozygous for the *Xrp1*$^{UAS\text{-}short}$ allele show modest background *Xrp1* activation and cell death even in the absence of Gal4.

(B, C) Co-expression of *Xrp1*$^{UAS\text{-}short}$ together with RFP (red) using the nub-Gal4 driver, induces Xrp1-HA (B) and cell death (Dcp1, in C).

(D-P) In experiments shown in panels D-P the act>y+>Gal4 cassette was used to generate clones; this cassette is weaker than the act>CD2>Gal4 cassette.

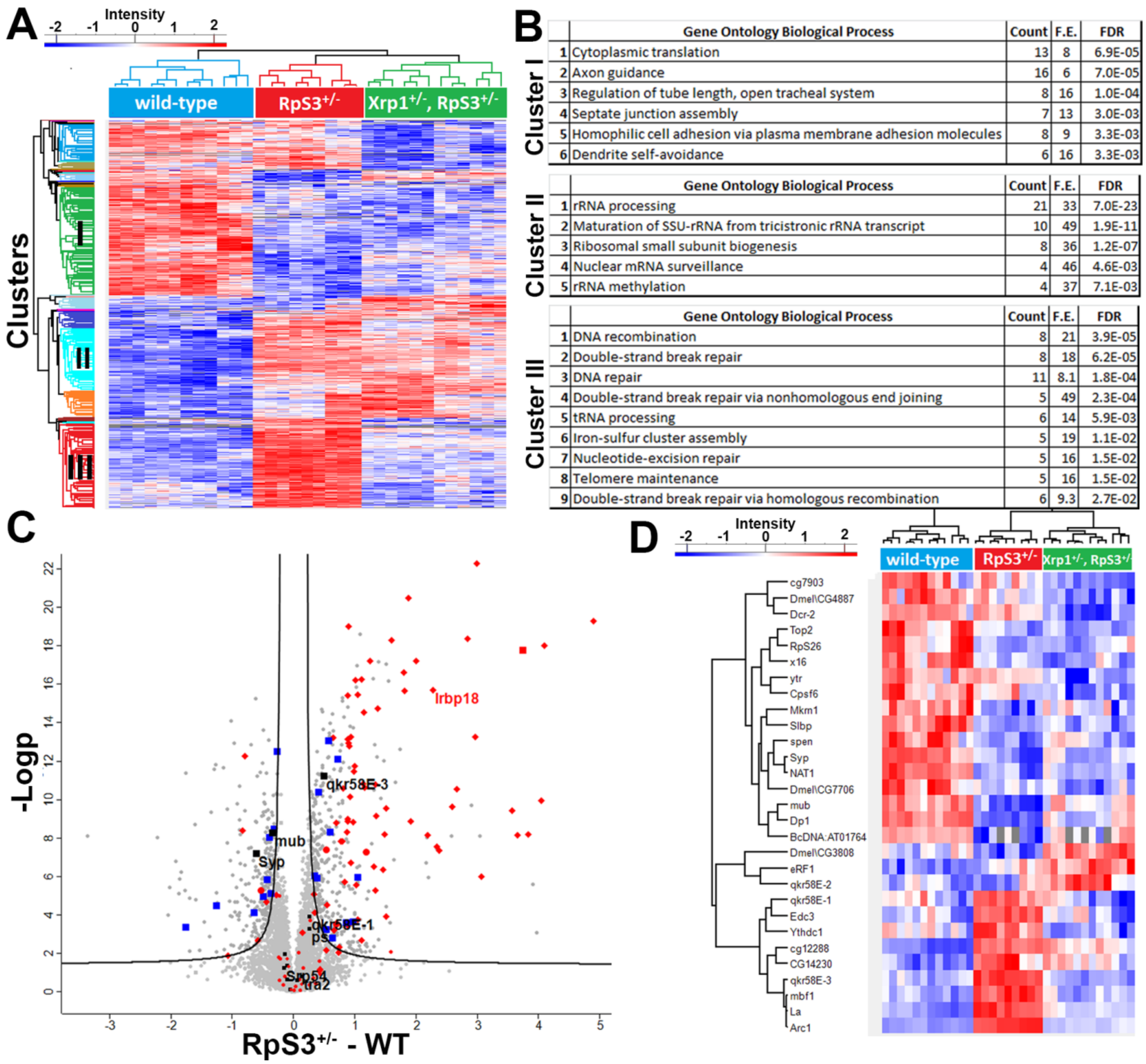
(D-L) Multiple examples of wing discs heterozygous with the *Xrp1*^{KO} allele, where *Xrp1*$^{UAS\text{-}short}$ is induced in clones (marked green by UAS-GFP) and promotes boundary cell death (Dcp1, magenta). 20 min heat-shock induced clones.

(M) Control discs expressing only GFP and w^{RNAi} in clones do not present increased cell death (Dcp1). 20 min heat-shock induced clones.

(N) Quantification of cell death per area in clones overexpressing *Xrp1*$^{UAS\text{-}short}$ compared to GFP clones from the experiments above. Each dot corresponds to single clones. For quantifications we used 11 wing discs for Xrp1$^{UAS\text{-}short}$ and n=7 wing discs for control. Statistical significance is shown as follows: ***$p < 0.001$; Wilcoxon rank sum test.

(O, P) Clones overexpressing Xrp1, using the homozygous *Xrp1*$^{UAS\text{-}short}$ allele, display prominent boundary cell death (Dcp1, magenta) in both wing (O) and eye imaginal discs (P). Clones were induced by a 40-min heat shock.

Figure S7. Related to Figure 6



Cluster I

	Gene Ontology Biological Process	Count	F.E.	FDR
1	Cytoplasmic translation	13	8	6.9E-05
2	Axon guidance	16	6	7.0E-05
3	Regulation of tube length, open tracheal system	8	16	1.0E-04
4	Septate junction assembly	7	13	3.0E-03
5	Homophilic cell adhesion via plasma membrane adhesion molecules	8	9	3.3E-03
6	Dendrite self-avoidance	6	16	3.3E-03

Cluster II

	Gene Ontology Biological Process	Count	F.E.	FDR
1	rRNA processing	21	33	7.0E-23
2	Maturation of SSU-rRNA from tricistronic rRNA transcript	10	49	1.9E-11
3	Ribosomal small subunit biogenesis	8	36	1.2E-07
4	Nuclear mRNA surveillance	4	46	4.6E-03
5	rRNA methylation	4	37	7.1E-03

Cluster III

	Gene Ontology Biological Process	Count	F.E.	FDR
1	DNA recombination	8	21	3.9E-05
2	Double-strand break repair	8	18	6.2E-05
3	DNA repair	11	8.1	1.8E-04
4	Double-strand break repair via nonhomologous end joining	5	49	2.3E-04
5	tRNA processing	6	14	5.9E-03
6	Iron-sulfur cluster assembly	5	19	1.1E-02
7	Nucleotide-excision repair	5	16	1.5E-02
8	Telomere maintenance	5	16	1.5E-02
9	Double-strand break repair via homologous recombination	6	9.3	2.7E-02

Figure S7 Proteomic analysis reveals Xrp1-dependent and independent changes of *Rp*$^{+/-}$ cells, including alterations in RNA-binding proteins.

(A) Hierarchical cluster analysis of differential expressed proteins between *wild-type*, *RpS3*$^{+/-}$ and *Xrp1*$^{+/-}$, *RpS3*$^{+/-}$ samples. 4 independent biological samples with 3 technical replicates were analyzed for *wild-type*, and *Xrp1*$^{+/-}$, *RpS3*$^{+/-}$ genotypes, while 3 independent biological samples with 3 technical replicates were analyzed for the *RpS3*$^{+/-}$ genotype. Blue indicates gradually downregulated proteins and red gradually upregulated proteins. Three selected clusters presenting distinct behavior with respect to proteomic changes are highlighted in the dendrogram. Cluster I*:* Xrp1-independent downregulated proteins in *Rp*$^{+/-}$ cells, Cluster II*:* Xrp1-independent upregulated proteins in *Rp*$^{+/-}$ cells; Cluster III*:* Xrp1-dependent upregulated proteins in *Rp*$^{+/-}$ cells.

(B) Functional analysis of protein annotation terms resulted in multiple categories of biological processes that were enriched in the three clusters. The enriched terms and the corresponding enrichment factor and *p*-value are shown. Cluster I comprised 232 proteins downregulated in *RpS3*$^{+/-}$ cells compared to wild-type, independent of Xrp1. This cluster was enriched for Gene Ontology terms associated with cytoplasmic translation, axon guidance, septate junction assembly, and dendrite self-avoidance. Cluster II included 134 proteins upregulated in RpS3$^{+/-}$ cells relative to wild-type, also independent of Xrp1. This cluster was enriched for terms related to ribosomal RNA processing and ribosome biogenesis, confirming previous findings showing that ribosome biogenesis intermediates accumulate independently of Xrp1 in Rp$^{+/-}$ cells[1,3]. These proteins in Clusters I and II represent potential candidates contributing to RpS12-dependent splicing of Xrp1. Finally, cluster III comprised 180 proteins upregulated in RpS3$^{+/-}$ cells in an Xrp1-dependent manner. This cluster showed enrichment for pathways involved in DNA recombination, DNA repair, DNA damage, and iron-sulfur cluster assembly. This confirms previous RNA-seq data showing that Xrp1 transcriptionally activates these pathways in *Rp*$^{+/-}$ cells[1].

(C) Volcano plot showing results of differential expression analysis comparing proteins from wild-type wing discs and *RpS3*$^{+/-}$ mutant wing discs. The vertical line differentiates the upregulated and the downregulated proteins. Square blue symbols indicate mRNA-binding proteins differentially expressed in *RpS3*$^{+/-}$ cells. Square black symbols indicate RBP previously shown to interact with Xrp1 transcript [8]. Diamond red symbols indicate differentially expressed proteins in *Rp*$^{+/-}$ cells, whose mRNA was also differentially expressed in *Rp*$^{+/-}$ cells[1]. Almost all of them belong to Cluster III. The presence of Irbp18 in Cluster III, further validates our proteomic analysis (Figure 6C). Note that less than 250 genes were differentially expressed based on our previous RNAseq analysis[1]. (x axis: log2 fold change, y axis: −log10 statistical p value) (FDR=0.01)

(D) Heatmap showing significantly altered RNA-binding proteins between *wild-type* wing discs, *RpS3*$^{+/-}$ discs and *Xrp1*$^{+/-}$,*RpS3*$^{+/-}$ discs. Blue indicates gradually downregulated proteins and red gradually upregulated proteins. Other RBPs that bind the same Xrp1 intron as Syncrip, including mub and qkr58E-1, are differentially regulated in *Rp*$^{+/-}$ cells.

Figure S8. Related to Figure 6

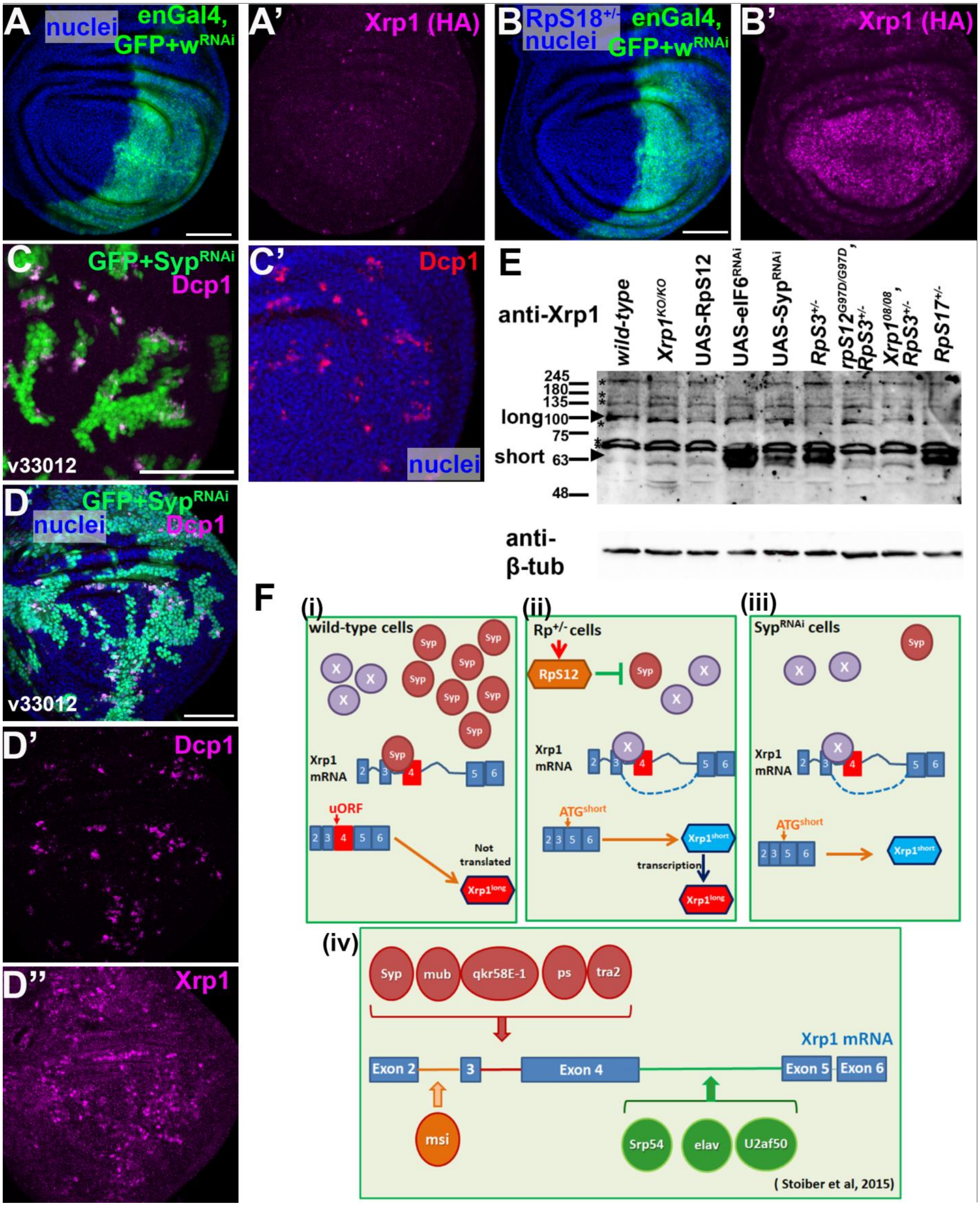


Figure S8. Syp depletion and Rp mutations induce Xrp1^{short} activation and cell death in wing discs.

(A-D) Confocal images of third instar wing discs of the indicated genotypes. Panels A and B show single confocal planes, while C and D show projections of the central disc-proper region. Nuclei are in blue. Scale bars, 50µm.

(A, B) en-Gal4 drives expression of GFP (green) together with w-RNAi in the posterior domain of wild-type (A) or *RpS18*$^{+/-}$ (B) wing discs. Anti-HA staining shows Xrp1-HA expression from the endogenously HA-tagged *Xrp1* allele. w-RNAi has no effect on Xrp1-HA levels (magenta) and serves as a negative control for Figure 6A-6D. Note that Xrp1-HA expression in *RpS18*$^{+/-}$ cells is not uniform across the disc and appears reduced at the dorsoventral boundary.

(C) Clonal *Syp* depletion by the weaker RNAi line (v33012) (green) in a wild-type disc induces cell death (active Dcp1 staining, magenta in C and red in C') at clone boundaries. Clones were induced by a 25-min heat-shock.

(D) Clonal *Syp* depletion by the weaker RNAi line (v33012) (green) in a wild-type disc strongly induces Xrp1 (GP anti-Xrp1 staining, magenta in D'') and boundary cell death (active cDcp1 staining, magenta in D'). Clones were induced by a 45-min heat-shock.

(E) Western blot of wing discs lysates from the indicated genotypes; part of this blot is shown in Figure 6G. Lanes: (1) *wild-type*, (2) Xrp1$^{KO/KO}$, (3) *actGal4, UAS-RpS12,* (4) *act>CD2>Gal4, UAS-eIF6RNAi,* (5) *act>CD2>Gal4, UAS-SypRNAi,* (6) *RpS3*$^{+/-}$, (7) *RpS12*$^{G97D/G97D}$*, RpS3*$^{+/-}$, (8) *Xrp1*$^{08/08}$*, RpS3*$^{+/-}$, and (9) *RpS17*$^{+/-}$. Xrp1^{short} isoform is strongly detected in eIF6-depleted cells, in Syp-depleted cells, in *RpS3*$^{+/-}$ and in *RpS17*$^{+/-}$ cells. Xrp1^{short} induction in *Rp*$^{+/-}$ cells is strongly suppressed by homozygosity of the *Xrp1*^{08} allele and of the *rpS12*^{G97D} allele. The Xrp^{long} isoform is barely detected in wild-type cells in the discs containing clones of eIF6-depleted or Syp-depleted cells. Xrp1^{long} protein appears strongly reduced in lane 3 *(actGal4, UAS-RpS12* cells), and in *RpS3*$^{+/-}$ cells. Note the background band just below the Xrp1^{long} band, which is also present in Xrp1 homozygous deletions. Background bands are marked with asterisks. Tubulin serves as loading control.

(F) Schematic models:

(i) In wild-type cells, Syncrip may negatively regulate $Xrp1^{short}$ expression by preventing the binding of another RBP that would otherwise promote exon 4 skipping. Translation of the abundant *Xrp1*^{long} RA/RG transcripts is likely reduced due by uORFs in their 5' UTR.

(ii) In *Rp*$^{+/-}$ cells, RpS12 likely reduces Syncrip (Syp) levels, either through specialized translation or an extraribosomal function. This reduction allows a competing RBP (X) to promote Xrp1 exon 4 skipping, producing the $Xrp1^{short}$ RC transcript, which is efficiently translated. The $Xrp1^{short}$ protein isoform then increases the levels of both isoforms by autoregulation,.

(iii) Syp depletion in wild-type cells mimics the *Minute* condition, allowing a competing RBP (X) to promote exon 4 skipping and produce the *Xrp1*^{short} RC transcript, which is then efficiently translated.

(iv) Schematic of the Xrp1 transcript showing the highest affinity binding sites of RNA-binding proteins identified by Stoiber et al. [8].

Figure S9. Related to Figure 7

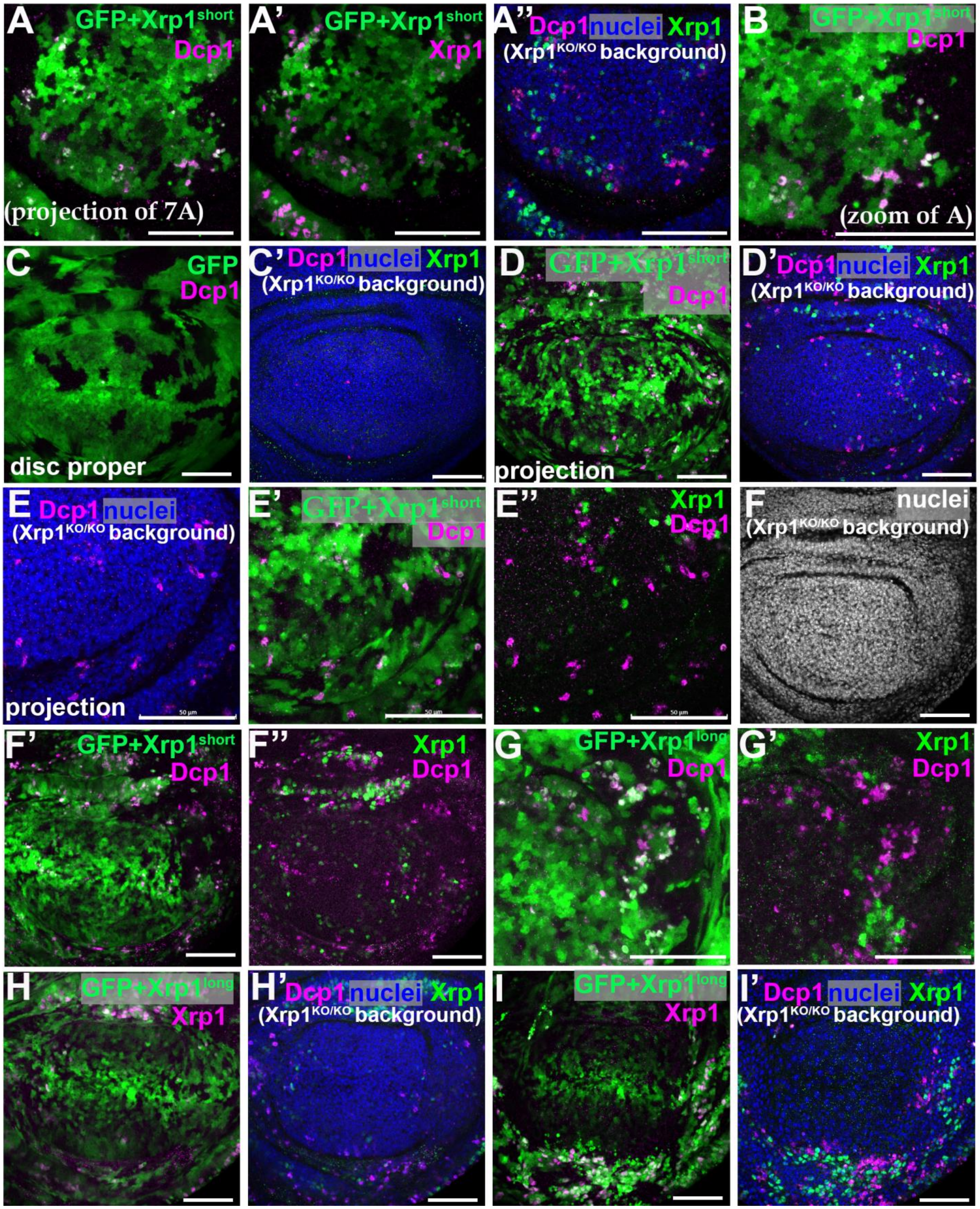


Figure S9. Boundary cell death and heterogeneous Xrp1 expression upon Xrp1 overexpression in wing mosaics.

Confocal images of third-instar wing discs of the indicated genotypes are shown. Clones were induced for 3 days in Xrp1$^{KO/KO}$ wing discs, using the act>y^{+}>Gal4 “weak” cassette and a 45-min heat shock. Panels A, B, D, E, G show projections of the central disc proper regions. All other panels show single confocal planes. Scale bars, 50μm.

(A, B) Higher magnification and projection of the disc shown in Figure 7A. Xrp1^{short} overexpressing clones (green) in Xrp1$^{KO/KO}$ wing discs exhibit boundary cell death (cDcp1, magenta in A and A’’) with non-uniform Xrp1 expression (magenta in A’ and green in A’’). Xrp1 levels are elevated at clone boundaries next to cDcp1-positive cells and reduced in clone centers. B shows a higher-magnification view of A, highlighting that cDcp1-positive cells are located predominantly at clone boundaries.

(C) Control GFP-overexpressing clones (green) in Xrp1$^{KO/KO}$ wing discs do not present cell death (cDcp1, magenta) or Xrp1 expression (green in C’).

(D, E, F) Two additional examples in which Xrp1^{short}-overexpressing cells (GFP, green) in the disc proper display sporadic Xrp1 expression (Xrp1, green in D’,E’’, F’’, and G’) and cDcp1-positive cells (magenta), located predominantly at clone boundaries. E shows a higher-magnification view of D, where it is evident that cells with higher Xrp1 are close to cDcp1-positive cells.

(G) Higher magnification of Figure 7D, highlighting that cDcp1-positive cells are located predominantly at clone boundaries. Xrp1^{long} overexpressing clones (green) in Xrp1$^{KO/KO}$ wing discs exhibit boundary cell death (cDcp1, magenta) with non-uniform Xrp1 expression (green in G’). Xrp1 is elevated at clone boundaries next to cDcp1-positive cells (magenta) and reduced in clone centers.

(H, I) Same disc as in Figure S5L-M, additionally showing Xrp1 expression. (H) Clonal overexpression of the Xrp1^{long} isoform (GFP) in a homozygous Xrp1^{KO} disc exhibits sporadic boundary cell death (cDcp1) and Xrp1 expression in the disc proper. (I) In the peripodial membrane of the same disc, Xrp1^{long}-overexpressing clones show prominent boundary cell death (cDcp1) and higher number of Xrp1-positive cells. The cells with high Xrp1 are not the cDcp1-positive cells, but are located adjacent to them.

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